



International Workshop on Photonics and Optical Materials (PHOTOMAT - 2026)



Organized by

Dept. of Applied Physics,
Defence Institute of
Advanced Technology,
Girinagar, Pune - 411025,
MH, India



October 7th - 9th, 2026



Takshshila Auditorium

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WORKSHOP SCHEDULE

International Workshop on Photonics and Optical Materials (PHOTOMAT-2026)

All times are in IST (+5:30 GMT)

Day 1: PHOTONICS (07 October 2026)		
Time	Training Contents	Lecturer/Instructor
9:00 am – 9:30 am	Registration and Inauguration	
9:30 am – 10:30 am	Plenary Lecture	Dr. Jagannath Nayak, Director General, MSS, DRDO
10.30 am – 10:45 am	High Tea Break	
10:45 am – 11:45 am	Design and Development of Ultrafast Yb-doped Fiber Oscillators and Amplifiers in Diverse Operating Regimes	Dr. P. K. Mukhopadhyay, SO-H, ALDAD, RRCAT.
11:45 am – 12:15 pm	All-Fiber Ultrafast Laser System for DEW Applications	Dr. Shyamal Mondal, Asst. Prof., Dept. of AP, DIAT, Pune
12.15 pm – 12.45 pm	Physical Demonstration by Industry	Revine Tech (Accelerating Innovation)
1:00 pm – 1:30 pm	Laboratory Session	Dr. A.V.R. Murthy, Dr. Shyamal Mondal, Mr. Pramod Bankar, Mr. Vikrant Gurmitkal
1:30 pm – 2:30 pm	Lunch Break	
2:30 pm – 5:00 pm	Laboratory Session	Dr. A.V.R. Murthy, Dr. Shyamal Mondal, Mr. Pramod Bankar, Mr. Vikrant Gurmitkal
5:00 pm – 6:00 pm	Optical Communication Technologies from Classical to Quantum: Challenges and Roadmap	Prof. Shailendra K. Varshney, Electronics and Electrical Communication Engg., IIT Kharagpur
6:00 pm – 7:00 pm	Building a long-distance quantum repeater network using atomic quantum Interconnects	Prof. Eden Valentin Figueroa Barragan, Director of Stony Brook, Quantum Institute, NY, USA
7:30 pm – 9:00 pm	Gala Dinner	
Day 2: OPTICAL WIRELESS COMMUNICATION AND UNDERWATER TESTING (08 October 2026)		
9:00 am – 10:00 am	Emerging Paradigms in Inertial Sensing: Overcoming Limits via Quantum Photonics	Sree Ramana Mukkamala Scientist H, DRDO-RCI, Hyderabad



10:00 am – 10:45 am	Cooperative Source Localization of Underwater Plumes using a Team of UUVs	Dr. Rakesh R. Warriar, Asst. Prof., NIT Calicut
10:45 am – 11:00 am	Tea Break	
11:00 am – 12:00 pm	Photonic Techniques for Generation, Distribution, and Measurement of Microwave Signals	Prof. Deepa Venkitesh, Dept. of EE, IIT Madras
12:00 pm – 12:30 pm	Underwater Vision Based Surveillance and Tracking using Machine Learning	Dr. Pooja Agarwal, Asst. Prof. Dept. of ME, DIAT, Pune
12:30 pm – 1:00 pm	Laser-Induced Photoacoustic Underwater Communication at 1 kbps Using a 1080 nm High- Power Fiber Laser	Prof. Devnath Dhirhe, Dept. of AP, DIAT, Pune
1:00 pm – 1:30 pm	Underwater Optical Communication System Design from Basic Building Blocks to Prototype	Dr. A.V.R. Murthy, Asst. Prof., Dept. of AP, DIAT, Pune
1:30 pm – 2:30 pm	Lunch Break	
2:30 pm – 6:30 pm	Laboratory Session	Dr. A.V.R. Murthy, Dr. Shyamal Mondal, Mr. Pramod Bankar, Mr. Vikrant Gurmitkal
Day 3: OPTICAL MATERIALS (09 October 2026)		
9:00 am – 10:00 am	The Organic-Inorganic Hybrid and Halide Perovskites: The Highly Tuneable New Breed of Optical Materials	Prof. Satishchandra B. Ogale, Director, RISE, TCG-CREST, Kolkata
10:00 am – 10:45 am	Terahertz Metasurface: A viable platform for exploring bound states in the continuum	Prof. Dibakar Roy Chowdhury, Anurag University, Hyderabad
10:45 am – 11:00 am	High Tea Break	
11:00 am – 11:45 am	Harnessing optical nonlinearities for ultrashort laser pulse applications	Dr. Aditya Dharmadhikari, Sc. Officer F, DNAP, TIFR, Mumbai
11:45 am – 12:15 pm	Carrier Injection as a versatile mechanism for optical switching and beam steering	Dr. Sooraj Ravindran, Asso. Prof., IIST, Trivandrum



12:15 pm – 1:00 pm	Innovative Design and Fabrication of Specialty Fibers for Enhanced Fiber Laser Performance	Dr. Anirban Dhar, Principal Scientist, FOPD, CGCRI, Kolkata
1:00 pm – 1:30 pm	Halide Perovskite Heterostructures for Photodetector Applications	Prof. Tejashree Bhawe, HoD, Dept. of AP, DIAT, Pune
1:30 pm – 2:30 pm	Lunch Break	
2:30 pm – 3:00 pm	Approaches at the interface of materials and optics: sensors and device development	Sr. Prof. Sangeeta Kale, Dept. of AP, DIAT, Pune
3:00 pm – 3:30 pm	Quartz-enhanced photo-acoustic spectroscopy (QEPAS) for detection of H ₂ S and NH ₃	Prof. Suwarna Datar, Dept. of AP, DIAT, Pune
3:30 pm – 4:00 pm	Valedictory Session	
4:00 pm – 4:30 pm	Evening Tea & Snacks	

Chief Guest

Dr. Jagannath Nayak
Director General, MSS, DRDO

Guest of Honor

Dr. Gaurav Singhal
Director, CHESS, DRDO Hyderabad

Chair Person

Prof. Tejashree Bhawe
HoD & Professor, DIAT

Convenor

Dr. Shyamal Mondal
Asst. Professor, DIAT

Co-Convenor

Dr. A V R Murthy
Asst. Professor, DIAT

Organizing Committee

Prof. Tejashree Bhawe	HoD & Professor, DIAT
Prof. Sangeeta Kale	Sr. Professor, DIAT
Prof. Suwarna Datar	Professor, DIAT
Prof. Devnath Dhirhe	Professor, DIAT
Dr. Shyamal Mondal	Asst. Professor, DIAT
Dr. A V R Murthy	Asst. Professor, DIAT
Dr. Pooja Agarwal	Asst. Professor, DIAT

MEET THE SPEAKERS



Dr. P. K. Mukhopadhyay

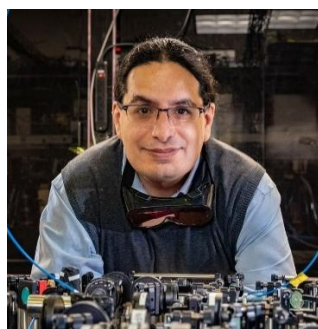
Scientific Officer – H,
Advanced Laser Development & Applications Division,
Raja Ramanna Centre for Advanced Technology, Indore

Bio: Dr. P. K. Mukhopadhyay is a distinguished scientist and researcher in the field of laser science and nonlinear optics. He is associated with the Solid State Laser Section, Advanced Laser Development and Applications Division (ALDAD), Raja Ramanna Centre for Advanced Technology (RRCAT), Indore. Dr. Mukhopadhyay obtained his Ph.D. in Physics from Devi Ahilya Vishwavidyalaya (DAVV), Indore, following an M.Sc. in Applied Physics from the University of Calcutta and a B.Sc. (Honours) in Physics from the University of Calcutta. His research interests encompass diode-pumped solid-state lasers, nonlinear optics, ultrafast fiber lasers and amplifiers, as well as laser applications including Laser-Induced Breakdown Spectroscopy (LIBS) and materials processing. His work spans both fundamental and applied aspects of laser technology, with particular emphasis on the development and application of advanced laser systems. Dr. Mukhopadhyay has also gained international research experience, having served as a Postdoctoral Scientist at Bilkent University, Turkey, during 2007–2008. His contributions have been recognized through several awards, including the DAE Science and Technology Excellence Award (2011) and DAE Group Achievement Awards (2009 and 2018). He has an extensive publication record, with around 70 publications in international journals, more than 100 conference papers, and four international book chapters. Through his research contributions in solid-state lasers, ultrafast photonics, nonlinear optics, and laser applications, Dr. Mukhopadhyay has made significant contributions to the development of advanced laser technologies in India.

Title of the talk: Design and Development of Ultrafast Yb-doped Fiber Oscillators and Amplifiers in Diverse Operating Regimes

Abstract: Train of ultrashort pulses with sufficient energy is required for numerous applications like in industrial material processing, aerospace, biomedical, defence and other branches of science and technology. In recent times, mode locked ytterbium (Yb) doped fiber laser (MLYbFL) and amplifiers are rapidly emerging as an attractive alternative to the solid-state counter part due to some inherent advantages like self-starting of modelocking process, flexibility, diffraction limited beam quality, high efficiency, easy thermal management and reliable misalignment free turn-key operation. However, due to very small core diameter and long length of the fiber the effects of group

velocity dispersion (GVD) and nonlinearity in fiber become very strong and their interplay leads to pulse distortion and even pulse breaking which in turn sets a limit on energy scaling from such system. To overcome these challenges ultrafast fiber laser has evolved through several modelocking regimes. The first MLYbFL was in dispersion managed configuration with net zero GVD, produced ultrashort pulses of sub 50fs duration with very low energy. With increase in GVD to net-normal generating similariton pulses and then to all-normal (ANDi) the oscillator producing dissipative solitons exhibits promise in energy scaling but at the cost of pulse duration. Recently, MLYbFL in Mamyshev oscillator (MO) configuration produced high energy pulses maintaining the ultrashort duration. However, it requires another laser to drive it and considerable efforts are going on to self-start the MO. In this talk we shall discuss the various regimes of modelocking in Yb-doped fiber laser to generate high energy ultrashort pulses with a highlight of our recent work. In this talk we shall also explore the functional possibilities of MLYbFL to generate pulses in diverse temporal format. Shaped pulses are generated by mode locking of Yb-doped fiber laser by implementing nonlinear loop mirror (NOLM) and nonlinear polarization rotation (NPR) based saturable absorber in a single cavity. The interplay of gain, dispersion, self phase modulation, NOLM and NPR generates a diverse mode-locking states producing pulses with diverse structures such as flat-top pulses with tunability in pulse duration in the range of 3ns- 100ns, step-like long pulses, bound-pulses, burst-mode pulses with adjustable number of short pulses in a burst etc. We also have generated pulse pattern with time varying properties like soliton rain, dual cavity modelocked laser, noise like pulses etc. Apart from bright pulses we also have generated dark pulses and dark-bright pulse pair from Yb-doped ANDi laser with novel configuration. The physical basis and state of the art of building these lasers with amplification characteristics of some of the lasers will be discussed in the talk.



Prof. Eden Valentin Figueroa Barragan
Director, Stony Brook Quantum Institute,
NY, USA

Bio: Prof. Eden Figueroa was awarded his BSc in Engineering Physics and his MSc in Optical Engineering at Monterrey Tech, Mexico in 2000 and 2002 respectively. From 2003 to 2008, he was a PhD student in the Quantum Technology Group of Prof. A. I. Lvovsky at the University of Konstanz in Germany and later at the Institute for Quantum Information Science at the University of Calgary, Canada. His PhD thesis entitled: “A quantum memory for squeezed light” was one of the first experimental implementations of quantum memory for quantized light fields. In 2009, he joined the Quantum Dynamics Group of Prof. G. Rempe at the Max-Planck-Institut für Quantenoptik in Garching, Germany where he worked in implementation of quantum networks utilizing single-atoms trapped in high-finesse optical cavities. Starting in 2013, he started the Quantum Information Technology Group at Stony Brook University. Currently he is Endowed Full Professor of Physics and the Director of the Stony Brook Quantum Institute. Since Jan.

2019, Prof. Figueroa is also a joint appointment with the Instrumentation Division at the Brookhaven National Laboratory.

Title of the talk: Building a long-distance quantum repeater network using atomic quantum Interconnects

Abstract: The goal of quantum communication is to transmit quantum states between distant sites. The key aspect to achieve this goal is the generation of entangled states over long distances. Such states can then be used to faithfully transfer classical and quantum states via quantum teleportation. The big challenge, however, is that the entanglement rates generated between two distant sites decreases exponentially with the length of the connecting channel. To overcome this difficulty, the new concepts of entanglement swapping, and quantum repeater operation are needed. In this talk we will show our progress towards building a quantum network of many quantum devices capable of distributing entanglement over long distances connecting Stony Brook University, the Brookhaven National Laboratory and other locations on Long Island, New York. We will show how to produce and communicate photonic quantum entanglement over a 275 km fiber network connecting five independent quantum nodes. Additionally, we will show our recent experiments in which we achieve Hong-Ou-Mandel interference across a 140 km, three-node quantum network. Lastly, we will discuss our progress towards building a memory-assisted quantum repeater network.



Prof. Satishchandra B. Ogale
Director,
RISE, TCG-CREST, Kolkata

Bio: Prof. Satishchandra Ogale is the Director of the Research Institute for Sustainable Energy (RISE, TCG-CREST), Kolkata, India as well as Adjunct Professor at the Indian Institute of Science Education and Research (IISER), Pune, India. Previously he was a faculty and Chair of the Physics Department at Pune University, Chief Scientist at CSIR-NCL, and Full Professor at IISER Pune. He has also worked at the University of Maryland at College Park from 1997 to 2006. Prof. Ogale has over 45 years of research experience in the field of Advanced Materials Science and his current research interests are in the field of Clean and Renewable Energy. He has over 550 research publications in reputed international peer-reviewed journals, 11 US Patents granted based on research done entirely in India, and three books co-edited for Springer, Wiley and Elsevier. His h-index is 95 with about 43000 citations. Prof. Ogale has won many National and International Awards and Recognitions. These include INSA Young Scientist Medal, N. S. Sathyamurthy award, B. M. Birla Prize, Sir C.V. Raman Prize, MRSI Medal, MRSI special silver jubilee Medal, NCL Scientist of the Year Prize, Ramanujan fellowship and Raja Ramanna fellowship. He is an elected fellow of the Indian Academy of Science and the National

Academy of Science. He is on the Editorial Advisory Boards of several high impact journals and has given numerous plenary and keynote talks in International and National Conferences. In 2022 he was won the “Researcher of the Year” award of the International Energy Storage Alliance (IESA) for his work on Energy Storage. More recently he has been declared winner of the prestigious UNESCO TWAS Award for Physics 2026, a Distinct Honour shared only by 16 Indian Physicists in the last 39 years. Prof. Ogale has won many National and International Awards and Recognitions. These include INSA Young Scientist Medal, N. S. Sathyamurthy award, B. M. Birla Prize, Sir C.V. Raman Prize, MRSI Medal, MRSI special silver jubilee Medal, NCL Scientist of the Year Prize, Ramanujan fellowship and Raja Ramanna fellowship. He is an elected fellow of the Indian Academy of Science and the National Academy of Science.

Title of the talk: The Organic-Inorganic Hybrid and Halide Perovskites: The Highly Tuneable New Breed of Optical Materials

Abstract: Photonics and optics are central to numerous application domains in the modern world. Across the whole region of the optical spectrum including UV, Visible, Near IR, and Far IR varied applications utilise the different facets of the interaction of photons with materials. Equally importantly, for the said applications, high intensity sources of photons need to be designed and implemented in the forms of lasers and light emitting diodes over these wavelength regimes. Towards this end, major advances have been achieved during the past decade with the emergence of a new series of optical materials belonging to the perovskite family, classified as Organic-Inorganic Hybrid Perovskites (OIHP) and Halide Perovskites (HP). The intriguing oxide perovskites with ABO_3 type structure have already delivered optical spectrum including UV, Visible, Near IR, and Far IR varied applications utilise the different facets of the interaction of photons with materials. Equally importantly, for the said applications, high intensity sources of photons need to be designed and implemented in the forms of lasers and light emitting diodes over these wavelength regimes. Towards this end, major advances have been achieved during the past decade with the emergence of a new series of optical materials belonging to the perovskite family, classified as Organic-Inorganic Hybrid Perovskites (OIHP) and Halide Perovskites (HP). The intriguing oxide perovskites with ABO_3 type structure have already delivered on numerous interesting phenomena and applications including high temperature superconductivity, piezoelectricity, colossal magnetoresistance and multiferroics. The new breed of hybrid and halide perovskites offer similar materials engineering opportunities covering optical and optoelectronic phenomena and devices. In this talk, I will outline this exciting field with a brief initial review followed by several examples derived from own research.



Sree Ramana Mukkamala
Scientist H,
DRDO-RCI, Hyderabad

Bio: Sree Ramana after obtaining his doctorate degree in Physics from University of Hyderabad had a brief stint of two years at Vasavi College of Engineering before joining RCI in 2005.

With a specialization in Modern Optics and Laser Physics, he contributed towards the development of laser-based sensors in the past 21 years and now exploring quantum technologies for sensing applications.

Title of the talk: Emerging Paradigms in Inertial Sensing: Overcoming Limits via Quantum Photonics

Abstract: This presentation provides an overview of current inertial sensor technologies, analysing their operational limitations across diverse environments. It highlights the critical demand for next-generation sensing capabilities, focusing on emerging technological frontiers. Special emphasis will be placed on quantum sensors that leverage laser and photonic systems. Finally, the session will explore the fundamental physical principles driving these advanced sensing modalities.



Dr. Rakesh Warriar R.
Assistant Professor,
Dept. of Electrical Engineering,
NIT, Calicut

Bio: Dr. Rakesh R Warriar is currently a faculty in the in the department of the Electrical Engineering. Before joining NIT Calicut, he worked as assistant professor in department of EEE in BITS Goa. He did his PhD from the department of Systems and Control Engineering, IIT Bombay. After his PhD, he did one year postdoc at the department of Mechanical and Aerospace engineering in Syracuse University, NY, USA. He has also worked as a postdoctoral researcher at the department of Aerospace Engineering in the Indian Institute of Science, Bangalore. His area of research is control systems with the focus on multi-agent systems. He also works on trajectory planning and control of unmanned aerial vehicles and underwater vehicles.

Title of the talk: Cooperative Source Localization of Underwater Plumes using a Team of UUVs

Abstract: Monitoring and tracking underwater plumes—whether of natural or anthropogenic origin—is crucial for environmental assessment, disaster mitigation, and marine ecosystem protection. This presentation proposes a cooperative framework for plume source localization using a network of unmanned underwater vehicles (UUVs) that measure a noisy scalar concentration field. A networked estimation strategy enables the UUVs to cooperatively estimate the unknown field gradient, guiding them toward the plume source. Results on cooperative estimation and tracking along under the communication constraints are discussed.



Prof. Deepa Venkitesh
Professor,
Dept. of Electrical Engineering,
IIT, Madras

Bio: Prof. Deepa Venkitesh obtained her Bachelors and Master's degree in Physics from the University of Kerala and PhD from the Indian Institute of Technology Bombay. She has worked in SAMEER Mumbai, VES College, Mumbai before joining the Indian Institute of Technology Madras. She is currently a Professor in the Department of Electrical Engineering, IIT Madras. She was bestowed with the Young Faculty Recognition Award of IIT Madras in 2012, and the Abdul Kalam Technology Innovation Fellowship of INAE in 2021. She is Senior Member of OSA, and Editorial Board member of the IET Optoelectronics. She was the Associate Vice President of "Women in Photonics" of the IEEE Photonics Society (2020-2023). She is also the National Coordinating Faculty for the Prime Ministers Research Fellowship program of the Govt of India. She is the project lead for the Advanced Optical Communication Testbed project- which is a consortium project funded by the Department of Telecommunications, Govt of India.

Title of the talk: Photonic Techniques for Generation, Distribution, and Measurement of Microwave Signals

Abstract: Microwave photonics addresses RF problems requiring bandwidth, frequency agility and low noise together, where electronic implementations become difficult. This talk reviews work built on a common set of photonic building blocks — low-phase-noise carrier generation, linear electro-optic conversion, dispersive time-frequency mapping, and optically timed sampling — reconfigured for different system functions: analog radio-over-fibre fronthaul, where digital transport bandwidth is traded against link linearity; optical phase-locked loops supplying coherent, agile local oscillators; photonic sampled ADCs replacing electronic aperture jitter with optical timing stability; dispersive and Talbot-based wideband spectrum sensing. Link loss, dynamic range and packaging, not bandwidth, remain the practical limits.



Prof. Dibakar Roy Chowdhury
Professor,
Dept. of Physics
Anurag University, Hyderabad



Bio: Dr. Dibakar Roy Chowdhury is currently Professor and Director of Research in Anurag University, Hyderabad, India. Prof. Roy Chowdhury pursued his PhD from Technical University of Darmstadt, Germany (2008). Later he worked at University of Duisburg-Essen (Germany), Los Alamos National Laboratory (USA), and Australian National University. His research interests are in the areas of Terahertz, metamaterials / metasurfaces, and quantum materials & devices. So far, he has published 125 papers in internationally reputed peer reviewed journals attracting more than 7200 citations with an h index of 39. He is a recipient of the prestigious and highly competitive SERB-STAR (Science and Technology Award for Research) award from the Government of India. He is currently a Fellow of the Institute of Physics (FInstP, UK). He is also an elected fellow of OPTICA (former OSA) for "innovations in passive and reconfigurable terahertz metamaterial devices and leadership in promoting terahertz activities in India".

Title of the talk: Terahertz Metasurface: A viable platform for exploring bound states in the continuum

Abstract: Bound states in the continuum (BIC) are discrete non-radiating eigenstates of a system that lie within the continuum of the radiation, characterized by the zero linewidth and theoretically infinite Q-factors¹. The BICs in photonics provide a unique pathway for the implementation of various applications such as sensing, lasing, detection, and non-linear enhancements². Symmetry-protected bound states in the continuum (SPBIC) are one of the BIC realization pathways widely explored in the terahertz (THz) metasurfaces due to their simpler yet exceptionally flexible implementation approaches. The SP-BICs arise due to the symmetry incompatibility of the state with respect to the incident radiation, thereby restricting the coupling of the mode to the radiation channel. Through symmetry perturbations of the system, introduced either by breaking structural symmetry of the system with respect to incident radiation or by asymmetric excitation of the system, these states can be accessed as leaky resonances with strong field confinements. Various unit cell geometries such as cut-wires, split ring resonators (SRR), and discs are utilized for the demonstration of SP-BICs in the THz metasurfaces. A metasurface design consisting of a pair of parallel cut-wires is employed to demonstrate the emergence of BIC from Fano resonance through symmetry restoration³. In another work, a pair of parallel SRRs are utilized to demonstrate the evolution of quasi-BIC resonances into BICs via dual excitation pathways. Here, the near-field coupling induced resonance mode hybridization of inductive-capacitive (LC) resonances leads to emergence of BIC for one probe polarization, while the near-field coupling of the dipolar resonances leads to emergence of BIC for orthogonal probe polarization⁴. These works highlight the emergence of SP-BICs and excitation of high Q-factor Fano-type resonances at the quasi-BIC conditions, providing enhanced light-matter interactions, which can pave the way for promising applications in the domains of THz functional devices and materials.



Dr. Aditya Dharmadhikari
Scientific Officer-F,
Dept of Nuclear and Atomic Physics
TIFR, Mumbai

Bio: Aditya K. Dharmadhikari is a Scientific Officer F in the Department of Nuclear and Atomic Physics at the Tata Institute of Fundamental Research, Mumbai, India. He obtained an M.Sc. (Physics) in 1990 and a PhD (Physics) in 1998 from the Department of Physics, University of Pune (currently Savitribai Phule Pune University). He worked on Lasers and Nonlinear Optics during his PhD. He was a Postdoctoral Fellow at Auburn University, USA (1998-2000). He then joined the Tata Institute of Fundamental Research as a Visiting Fellow in 2001. He has published more than 110 peer-reviewed papers and delivered 50 invited talks. His current research interests include ultrafast nonlinear optics and Terahertz generation.

Title of the talk: Harnessing optical nonlinearities for ultrashort laser pulse applications

Abstract: Ultrafast femtosecond lasers are crucial for photonics applications in both research and industry. They find utility in basic research such as strong-field physics, time-resolved spectroscopy, and broadband Terahertz generation. These lasers also have many industrial applications, such as microstructuring, precise material machining, processing lightweight materials, and manufacturing high-precision implants and medical devices. The nonlinear optical properties of transparent materials are crucial for generating a broad spectrum through self-phase modulation to reduce the pulse duration of ultrafast lasers. In this talk, I will discuss how optical nonlinearities of transparent solids and gases can be harnessed for various femtosecond laser-based applications.



Dr. Sooraj Ravindran
Associate Professor,
Dept. of Avionics,
IIST, Trivandrum

Bio: Dr. Sooraj Ravindran did B.Tech in Electronics and Communication engineering from NSS college of Engineering Palakkad under University of Calicut, M.Tech in Optoelectronics and Laser Technology from International School of Photonics, Cochin University of Science and Technology in 2001. After working at Research Center Imarat,



DRDO-Hyderabad on Integrated Optics for two years, he left for Republic of Korea where he did his PhD and Postdoctoral research at Gwangju Institute of Science and Technology, focusing on III-V material based microring resonator switches, electro-absorption modulators, optical interconnects and multi-junction solar cells. Since 2015, he is with Department of Avionics, Indian Institute of Space science and Technology (IIST), Trivandrum, and currently is an Associate Professor where his research activities focus on Semiconductor Optoelectronic devices, multi-junction solar cells, and microring resonator-based switches and modulators.

Title of the talk: Carrier Injection as a versatile mechanism for optical switching and beam steering

Abstract: The ability to dynamically control the flow of light is fundamental to a wide range of photonic technologies, including optical communication, photonic integrated circuits, LIDAR, and optical computing. Carrier injection provides a simple yet powerful mechanism for achieving such control by electrically modifying the optical properties (refractive index) of a semiconductor. In this talk, we explore how carrier-injection-induced refractive index modulation can be translated into different photonic functionalities, with particular emphasis on optical switching and beam steering. The talk will introduce the basic physics of carrier injection in semiconductor photonic devices and the resulting changes in refractive index and optical absorption through carrier-induced effects such as band filling, bandgap shrinkage, and free-carrier dispersion. These concepts are then illustrated through two device implementations. First, a GaAs-based microring resonator optical switch employing a p-i-n structure is discussed. Injection of electrons and holes into the intrinsic region through an electrical bias changes the refractive index and consequently shifts the resonance of the microring, enabling optical switching in the 1300 nm and 1550 nm telecommunication bands. This approach demonstrates the possibility of using GaAs for long-wavelength operation without relying on conventional quantum-well-based electroabsorptive structures. The talk will then extend the same carrier-injection principle to optical beam steering using a silicon-based optical phased array. Controlled carrier injection in PIN structures provides dynamic phase modulation and enables two-dimensional beam steering at a fixed wavelength of 1550 nm. Transverse and longitudinal steering are achieved through phase control and novel tunable grating antennas, respectively, avoiding the need for wavelength-tunable light sources. By connecting the underlying carrier physics with these two distinct device architectures, we will demonstrate the versatility of carrier injection as a mechanism for controlling light. The discussion will also address practical aspects such as modulation speed, power consumption, optical loss, scalability, and device integration, providing an overview of the opportunities and challenges associated with carrier-injection-based photonic devices.



Dr. Anirban Dhar
Principal Scientist,
Fiber Optics & Photonics Division,
CGCRI, Kolkata

Bio: Dr. Anirban Dhar is working as Scientist-F in the Fiber Optics and Photonics Division at CSIR CGCRI, Kolkata. He earned his Ph.D. from Jadavpur University in 2008 and completed postdoctoral research at the Institute of Photonics and Electronics (Czech Republic) and the Optoelectronics Research Center (UK) from 2009–2012 before joining CSIR-CGCRI. Dr. Dhar has made notable contributions to optical fiber technology, including leading the development of the world's first multimode amplifier, pioneering alternative dopants, and advancing fiber laser technology with active GRIN fibers. His research focuses on specialty optical fibers for amplifiers, high-power lasers, and sensors. He has published over 120 SCI journal papers, 80 conference papers, 4 U.S. patents, and 4 book chapters. He is also an Associate Professor at AcSIR, India, has delivered numerous invited talks globally, and currently mentors three Ph.D. students (with one already graduated). His accolades include the MRSI Young Scientist Award, OSA Senior Member recognition, and the DRDO DARE to DREAM award.

Title of the talk: Innovative Design and Fabrication of Specialty Fibers for Enhanced Fiber Laser Performance

Abstract: Ytterbium-doped fiber lasers operating around the 1-micron wavelength have emerged as the most reliable and widely adopted laser technology, thanks to their superior beam quality, high lasing efficiency, low thermal load, and compact form factor. These advantages have enabled them to nearly replace traditional solid-state lasers across a wide range of applications, including industry, medicine, scientific research, and strategic sectors. Currently, continuous-wave (CW) ytterbium fiber lasers have achieved power levels of approximately 20 kW with near-single-mode beam quality, while multimode systems have surpassed hundreds of kilowatts. Despite these achievements, ongoing research aims to push power scaling further while simultaneously addressing key limitations in fiber laser performance. Major challenges at high powers include nonlinear effects such as Stimulated Brillouin Scattering (SBS), Stimulated Raman Scattering (SRS), photodarkening, and transverse modal instabilities. One of the most effective strategies to mitigate these issues involves the development of active fibers with novel waveguide designs and tailored host glass compositions. This lecture will highlight the critical role of waveguide engineering and material composition in customizing and optimizing fiber laser performance, with a focus on overcoming the present limitations that arise at high power levels.



Prof. Shailendra K Varshney

Professor,
Electronics and Electrical Communication Engg.,
IIT Kharagpur

Bio: Dr. S. K. Varshney is recognized for his pioneering research in fiber optics and photonic technologies, encompassing integrated nanophotonics, metasurfaces, nonlinear optics, and quantum photonics. He is currently a professor in the Department of Electronics & Electrical Communication Engineering at IIT Kharagpur. He has made pioneering contributions to specialty fiber optics, microring resonators, linear and nonlinear devices, fiber-based sources, and optical and quantum communications. He is the recipient of several fellowships UGC/CSIR JRF, the Monbukagakusho scholarship from the Government of Japan, the JSPS fellowship from Japan, and the Alexander von Humboldt fellowship and DAAD fellowship from Germany. He is the recipient of the Faculty Excellence Award (2019) by IIT Kharagpur and the G S Sanyal Faculty Excellence Award (2020) by the IIT Kharagpur Alumni Foundation USA. He is a senior member of IEEE and Optica. He is the author and co-author of over 120 research papers in peer-reviewed journals and more than 150 papers in conference proceedings. He also holds 5 Indian patents to his credit.

Title of the talk: Optical Communication Technologies from Classical to Quantum: Challenges and Roadmap

Abstract: In this talk, fundamentals of various optical communication technologies (fiber-optic, free-space, underwater) will be discussed, covering quantum aspects as well. The current challenges and the roadmap will also be discussed.



Dr. Pooja Agarwal

Assistant Professor,
Dept. of Mechanical Engineering,
DIAT, Pune

Bio: Dr. Pooja Agarwal is currently an Assistant Professor at DIAT (Defence Institute of Advanced Technology) in Pune. She completed her Ph.D. from the Indian Institute of Science (IISc), Bangalore, focusing on the automation and intelligence of advanced machinery. Dr. Agrawal serves as the primary Laboratory-in-charge for the institute's specialized Advanced Robotics Lab. Her high-impact engineering research specifically

covers the guidance, automated control, and motion planning of Autonomous Vehicles, Unmanned Aerial Vehicles (UAVs), and robot vision systems.

Title of the talk: Underwater Vision Based Surveillance and Tracking using Machine Learning

Abstract: Underwater environments present significant challenges for vision-based surveillance and tracking due to turbidity, poor illumination, light scattering, and severe color distortion. These conditions degrade image quality and make it difficult for conventional vision systems to reliably detect objects and continuously track their movement. This talk explores how machine learning, underwater image enhancement, object detection, and tracking techniques can be integrated to improve visual perception in challenging underwater environments. The talk presents ModYOLO, a practical enhancement-aware framework that combines underwater image enhancement with machine-learning-based object detection to improve the visibility and identification of underwater objects. The system development encompasses underwater data collection, dataset preparation, model training, evaluation, and real-time deployment. Experiments conducted using data from three diverse environments—Aquarium, Dam, and Well—demonstrate the effectiveness of the approach under different visibility and scene conditions. Beyond object detection, the talk focuses on vision-based tracking, where detected objects can be monitored across successive video frames to estimate their movement and support continuous underwater surveillance. This combination of detection and tracking provides a more comprehensive understanding of dynamic underwater scenes than frame-by-frame object identification alone. The developed system also incorporates an interactive interface to support real-time monitoring and practical deployment. The presentation will highlight the role of machine learning in enabling reliable underwater object detection and tracking, and discuss its potential applications in underwater inspection, environmental monitoring, search and surveillance, marine robotics, and autonomous underwater systems.



Dr. Shyamal Mondal
Assistant Professor,
Dept. of Applied Physics,
DIAT, Pune

Bio: Dr. Shyamal Mondal, is working as an Assistant Professor at the Department of Applied Physics, Defence Institute of Advanced Technology, Pune, India. His broad research areas are designing and developing ultrafast fiber and solid-state laser oscillators and amplifiers, High-electric-field THz generation and detection, nonlinear optics and their applications. He completed his Ph. D. from IIT Kharagpur in 2015 with a specialization in 'Ultrafast Lasers and Nonlinear Optics'. He worked as a Rutherford International Fellow at STFC-Daresbury Laboratory, UK, and as a Postdoctoral Fellow at the Institute of Photonics and Electronics, Prague, Czech Republic. He has completed two (02) high-value projects funded by DIAT and DRDO. He has published more than 60 peer-

reviewed articles and three book chapters. He is a regular reviewer of Elsevier, Optica, Wiley, etc., and life member of OSI, ILA and a Young Scientist member of OPTICA.

Title of the talk: All-Fiber Ultrafast Laser System for DEW Applications

Abstract: Directed Energy Weapons (DEWs) are emerging as an important technology for next-generation defence systems, offering the potential for precision engagement, rapid response, and reduced dependence on conventional interceptors. For India, developing indigenous high-power and ultrafast laser sources is strategically relevant to future applications in counter-UAS, electro-optical systems, precision sensing, and other directed-energy technologies. In this context, all-fiber ultrafast laser systems offer significant advantages in terms of compactness, alignment-free operation, high efficiency, thermal management, scalability, and ruggedness compared with conventional bulk-optic laser architectures.

This talk presents the principles, architecture, and technological potential of an All-Fiber Ultrafast Laser System for Directed Energy Weapon (DEW) Applications. The discussion covers ultrashort-pulse generation, nonlinear pulse evolution, fiber-based amplification, coherent and spectral control, pulse compression, and power scaling. Particular emphasis is placed on overcoming key challenges associated with nonlinear effects, amplified spontaneous emission, thermal loading, optical damage, and beam-quality preservation during power scaling, etc. The presentation will discuss advanced fiber architectures, large-mode-area fibers, photonic-crystal/hollow-core fibers, and nonlinear pulse-compression techniques as possible routes to higher peak power and improved beam quality.

It also highlights the relevance of indigenous photonic technologies and subsystem development for establishing a scalable Indian all-fiber ultrafast laser technology platform. Such a platform can contribute to the development of compact, efficient, and robust laser sources for future defence and aerospace applications, while also enabling dual-use capabilities in high-precision manufacturing, remote sensing, LiDAR, nonlinear spectroscopy, and scientific research.



Prof. Devnath Dhirhe
Professor,
Dept. of Applied Physics
DIAT, Pune

Bio: Dr. Devnath Dhirhe, is working as Professor at the Department of Applied Physics and Joint Registrar, Defence Institute of Advanced Technology, Pune, India. His broad research areas are MiD-IR and THz Quantum Cascade Laser, Quantum Dot Lasers, Integrated Optics and Si-Photonics, Integrated Polarisation Manipulation, Ring Lasers, Fiber Laser/Low Noise Fiber Amplifier, High Speed Free-Space Optical Communication, Telecommunication 1550 nm Semiconductor Laser. He has done his Ph.D from University of Glasgow, Glasgow, UK.

Title of the talk: Laser-Induced Photoacoustic Underwater Communication at 1 kbps Using a 1080 nm High-Power Fibre Laser

Abstract: This work presents an underwater communication system based on laser-induced photoacoustic technology using a high-power 1080 nm fiber laser. In this approach, modulated laser pulses interact with water to generate acoustic pressure waves through rapid thermoelastic expansion. These photoacoustic waves propagate through the water and are detected with an underwater hydrophone to recover information. The system is designed to achieve a data rate of 1 kbps in the laboratory, demonstrating the feasibility of laser-based acoustic communication for challenging underwater environments. The proposed technique offers non-contact transmission and reduced sensitivity to optical scattering, with potential applications in underwater surveillance, autonomous underwater vehicles, naval communication, and oceanographic sensing.



Dr. A. V. R. Murthy
Assistant Professor,
Dept. of Applied Physics,
DIAT, Pune

Bio: Dr. Murthy, is working as an Assistant Professor at the Department of Applied Physics, Defence Institute of Advanced Technology, Pune, India. His broad research areas are Free Space Optical Communication, Optical instrumentation, Bio-photonics, Atomic Force microscopy and spectroscopy of biomimetic systems. He has done his Ph.D from IISER pune and post doc from France. He was awarded DST-Inspire Faculty fellowship in 2015. He published more than 40 peer review articles.

Title of the talk: Underwater Optical Communication System Design from Basic Building Blocks to Prototype

Abstract: Underwater Optical Communication (UWOC) offers high-speed and low-latency data transmission for emerging underwater applications. This talk provides a practical overview of UWOC system development, beginning with the fundamental principles and selection of key building blocks, including the optical source, driver circuit, modulation scheme, receiver, photodetector, and data interface. The integration of these components into an end-to-end communication link is discussed, along with practical challenges such as optical alignment, attenuation, scattering, turbidity, and reliable data transmission. The talk further covers the transition from individual components to a functional laboratory prototype, demonstrating how theoretical concepts can be implemented and evaluated under real underwater conditions. The overall objective is to provide a clear understanding of the design considerations involved in developing a reliable and practical underwater optical communication system.



Prof. Tejashree Bhawe
Professor & HoD,
Dept. of Applied Physics
DIAT, Pune

Bio: Dr. Tejashree Bhawe received her PhD from the University of Pune (SPPU), Pune, Maharashtra, India in the field of Nano Silicon. She worked as a CSIR Research Associate in SPPU and as Post-Doctoral Fellow in Centre for Interdisciplinary Research, Tohoku University, Sendai, Japan. She also worked as a Visiting Scientist at INFN-LNF, Italy. Before joining DIAT she was working as Assistant Professor in Post Graduate Department of Electronic Science, SPPU. She has been working as an Associate Professor at the Department of Applied Physics, Defence Institute of Advanced Technology, Pune, India. Her broad research areas are Photovoltaics, Microfluidic devices, Nano materials applications for Sensors, Resistive switching etc. She has published 70 research articles in reputed scientific journals with 1177 citations, h-index 19 and i-10 index of 30.

Title of the talk: Halide Perovskite Heterostructures for Photodetector Applications

Abstract: Innovative material designs that improve light absorption, charge transfer, and device stability are essential to the development of high-performance optoelectronic devices. In this study, we investigate the design and manufacturing of heterostructure-based photodetectors through the integration of halide perovskites, quantum dots, and 2D materials. Hybrid devices are perfect for flexible electronic applications and broadband photodetection because they have tuneable optical and electronic characteristics^{1,2}. For better photodetection performance, three distinct heterostructures were created and investigated. On a flexible paper substrate, a MoS₂/N-GQDs/CsPbBr₃ triple layer junction photodetector was created, exhibiting effective charge separation and improved responsivity as a result of robust interfacial contacts. As a lead-free substitute, a Cs₂AgBiBr₆/MoS₂ heterostructure was created, which shows better charge transfer and lower recombination losses. Furthermore, a Cs₂AgBiBr₆/MoS₂ heterojunction photodetector demonstrated outstanding responsiveness, taking use of the complementary interaction between 2D MoS₂ nanosheets and 0D perovskite nanocrystals. These heterostructure-based devices show promise for flexible and wearable technology, as well as enhanced optoelectronic performance and stability. The paper emphasizes how interface optimization and material engineering are advancing next-generation photodetectors for use in transparent electronics, imaging, and sensing. In order to further maximize device performance, future research will concentrate on improving scalability, stability, and investigating novel material combinations.



Prof. Sangeeta N. Kale
Senior Professor,
Dept. of Applied Physics,
DIAT, Pune

Bio: Dr. Sangeeta Kale, graduated from University of Pune (SPPU), India and did her Masters in Electronic-Science and Doctoral studies in Material Science from the same university in 1996. She did her post-doctoral studies from University of Maryland, College Park, U.S.A., from 2000 to 2002. She has been working at Defence Institute of Advanced Technology as Professor and Director (Policy and Planning). Additionally, she has been a visiting Scientist at the International Centre for Theoretical Physics, since 2006, till date, at Trieste, Italy.

Title of the talk: Approaches at the interface of materials and optics: sensors and device development

Abstract: Current materials and photonics research centers on integrating novel low-dimensional materials—such as 2D graphene, wide band-gap semiconductors, and thin-films onto silicon platforms to dramatically scale energy efficiency, optical processing speeds, and data bandwidths. Concurrently, these engineered materials enable highly sensitive, ultra-compact photonic sensors capable of single-molecule biomedical diagnostics, environmental tracking, and integrated system development. Alongside, technology has significantly altered how medicine is delivered in general and surgery in particular during the last few decades. This shift in strategy has made less invasive diagnostic and treatment methods possible.

Through this presentation, few strategies are discussed using which optical properties of novel materials are used along with various interferometric modes of optical fibers to realize miniaturized sensors. Modes and type of fibers (normal and specialty fibers) along with stimuli-responsive optical materials together form smart sensors, which will be discussed. A technique for photoplethysmography (PPG) will be also evaluated to diagnose blood hemoglobin using an optical approach, which has several advantages over conventional pathological approach.



Prof. Suwarna Datar
Professor,
Dept. of Applied Physics,
DIAT, Pune



Bio: Dr. Suwarna S. Datar received her PhD from the University of Pune (SPPU), Pune, Maharashtra, India in the field of Scanning Probe Microscopy and Spectroscopy Techniques. She is currently a Professor and Dean (Training and Placements), Defence Institute of Advanced Technology (DIAT), an autonomous Deemed University under the Ministry of Defence. Her research interests are probe microscopy, graphene and its composites for EMI shielding, plasmon polariton and MEMS-based sensors, and breathomics.

Title of the talk: Quartz-enhanced photo-acoustic spectroscopy (QEPAS) for detection of H₂S and NH₃

Abstract: The detection of trace gases at parts-per-million (ppm) and sub-ppm concentrations is a prerequisite for a wide range of scientifically and industrially critical activities: environmental monitoring of atmospheric pollutants, industrial hazard mitigation in chemical and petroleum facilities, non-invasive medical diagnostics through exhaled breath analysis, and process control in semiconductor fabrication. One of the ways to detect these trace gases is through laser absorption spectroscopy with semiconductor diode lasers (~a few MHz line width) which can resolve individual vibrational absorption lines of gases which are fingerprints for detection

Despite the advantages of selectivity and dynamic range, direct absorption spectroscopy suffers from excess 1/f noise at low detection frequencies and requires long optical pathlengths (> 10 m in some cases) to achieve sub-ppm sensitivity in the near-infrared (NIR), where molecular line strengths are weaker than in the mid-infrared fundamental bands.

Quartz-enhanced photo-acoustic spectroscopy (QEPAS) was introduced by Kosterev et al. in 2002 as an elegant solution to the path-length limitation of direct absorption techniques. In QEPAS, the light absorbed by the target gas is converted into an acoustic wave through the process of non-radiative (vibrational-translational, V-T) relaxation. A standard 32.768 kHz quartz tuning fork (QTF), mass-produced for the watch industry at negligible unit cost, serves as the acoustic transducer. By modulating the optical excitation at the mechanical resonance frequency of the QTF, the photo-acoustic signal is amplified by the quality factor $Q \approx 10,000$ at atmospheric pressure, achieving sensitivities in the parts-per-billion range with centimeter-scale interaction lengths.

The practical utility of QEPAS has nonetheless been restricted by two architectural bottlenecks. First, the feeble piezoelectric current output of the QTF (typically 1–100 pA at ppm concentration levels) necessitates front-end transimpedance amplification followed by phase-sensitive lock-in demodulation which increases the cost. By fabricating a biconical single-mode fiber taper with a waist diameter of 13.6 μm , evanescent-wave excitation is delivered directly and self-aligned into the QTF gap, eliminating free-space optics. By implementing a software digital lock-in amplifier (D-LIA) on an Arduino Uno micro-controller, the cost of electronic demodulation is reduced from lakhs of rupees to under few hundreds. The complete EW-QEPAS sensor, capable of sequential detection of H₂S and NH₃, is realized in the present work. Working of QEPAS with tapered fiber and its application in detection of trace gases will be discussed.

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LABORATORY SESSION MANUAL

International Workshop on Photonics and Optical Materials (PHOTOMAT-2026)

Experiment 1: Green Synthesis of Quantum Dots via Pulsed Laser Ablation in Liquid

Objective:

To synthesize colloidal quantum dots (QDs) using Pulsed Laser Ablation in Liquid (PLA) and evaluate the resulting nanoparticle suspension, completely avoiding toxic chemical precursors.

Theory:

Pulsed Laser Ablation in Liquid is a "top-down" green synthesis method. When a high-power, focused laser pulse (typically from a Q-switched Nd:YAG laser) strikes a solid bulk target submerged in a liquid (such as deionized water or ethanol), the intense irradiance induces localized optical breakdown.

This rapidly generates a high-temperature, high-pressure plasma plume at the solid-liquid interface. As the plasma expands and interacts with the confining liquid, it cools at extreme rates, leading to the nucleation and growth of nanoparticles and quantum dots. The surrounding liquid acts as the ablation medium and, in many green protocols, the natural capping agent to prevent severe agglomeration.

Procedure:

- **Target Preparation:** Mechanically and chemically clean the solid bulk target to remove surface oxides. Ultrasonicate the target in ethanol for 10 minutes, followed by rinsing in deionized (DI) water.
- **Mounting:** Fix the target securely at the bottom of a clean quartz cuvette or glass beaker.
- **Solvent Addition:** Add a measured volume of green solvent (e.g., ultra-pure DI water) so the liquid level is approximately 5 to 10 mm above the target surface.
- **Optical Alignment:** Align a high-power pulsed laser beam (e.g., 1064 nm or 532 nm) perpendicularly to the target. Use a plano-convex lens to focus the beam slightly below the target surface to achieve the necessary fluence (J/cm^2) for optical breakdown without causing splashing.
- **Ablation:** Set the laser repetition rate (e.g., 10 Hz) and pulse energy. Initiate ablation for 15 to 30 minutes. To prevent deep crater formation which alters the focal spot, place the beaker on an automated X-Y translation stage.
- **Collection:** Carefully extract the colloidal suspension. Centrifuge the liquid at high speeds (e.g., 10,000 rpm for 15 mins) to precipitate larger unablated micro-particles. Extract the supernatant containing the quantum dots.

Expected Results:

The liquid will exhibit a distinct color change dependent on the target material and particle size (e.g., yellow/brown for certain carbon or metallic QDs). When illuminated with a UV excitation source, the suspension should display visible photoluminescence (PL), confirming the quantum confinement effects of the synthesized dots.

Precautions:

- ✓ **Laser Safety:** Strict adherence to laser safety protocols is mandatory. Wear appropriate optical density (OD) laser safety goggles matched to the emission wavelength. Ensure no stray specular reflections escape the setup
- ✓ The target must be firmly anchored; the shockwave from the plasma plume can dislodge loose targets, causing them to float into the beam path and scatter dangerous reflections.
- ✓ Carefully monitor the liquid level, as evaporation during high-power ablation changes the optical path length and focal spot size.



Experiment 2: Advanced Optoelectronic Characterization of LEDs

Objective:

To extract the fundamental electrical and optical parameters of Light Emitting Diodes (LEDs)—such as spectral power distribution, CRI, CCT, LER, R9 —using an integrating sphere and a Source-Measure Unit (SMU).

Theory:

Comprehensive LED characterization requires simultaneous mapping of electrical injection and optical emission. Electrically, an LED follows the standard diode equation, characterized by its forward turn-on voltage and parasitic series resistance. Optically, the device performance is defined by its L-I (Light-Current) curve.

Because LEDs possess highly divergent emission profiles, a standard photodiode cannot capture the total emitted flux. An integrating sphere—a hollow cavity coated with a highly diffuse reflective material (like Barium Sulfate)—spatially integrates the light emitted in all directions. A baffled exit port connects to a spectrometer to capture the spectral power distribution.

Procedure:

- **Calibration:** Turn on the standard calibration (halogen/tungsten) lamp inside the integrating sphere. Allow it to thermally stabilize for 15-20 minutes. Record the reference spectrum to calibrate the coupled spectrometer's responsivity.
- **Mounting:** Turn off the calibration lamp. Mount the test LED onto the center sample holder inside the integrating sphere. Ensure the baffle properly blocks direct line-of-sight between the LED and the detector port.
- **Electrical Connection:** Connect the LED to a precision Source-Measure Unit (SMU).
- **Data Acquisition Setup:** Interface the SMU and the spectrometer with a PC.
- **Measurement Sweep:** Command the SMU to perform a forward-bias current sweep (e.g., 0 to 100 mA in 5 mA steps). At each step, record the forward voltage, and trigger the spectrometer to record the emission spectrum.
- **Analysis:**

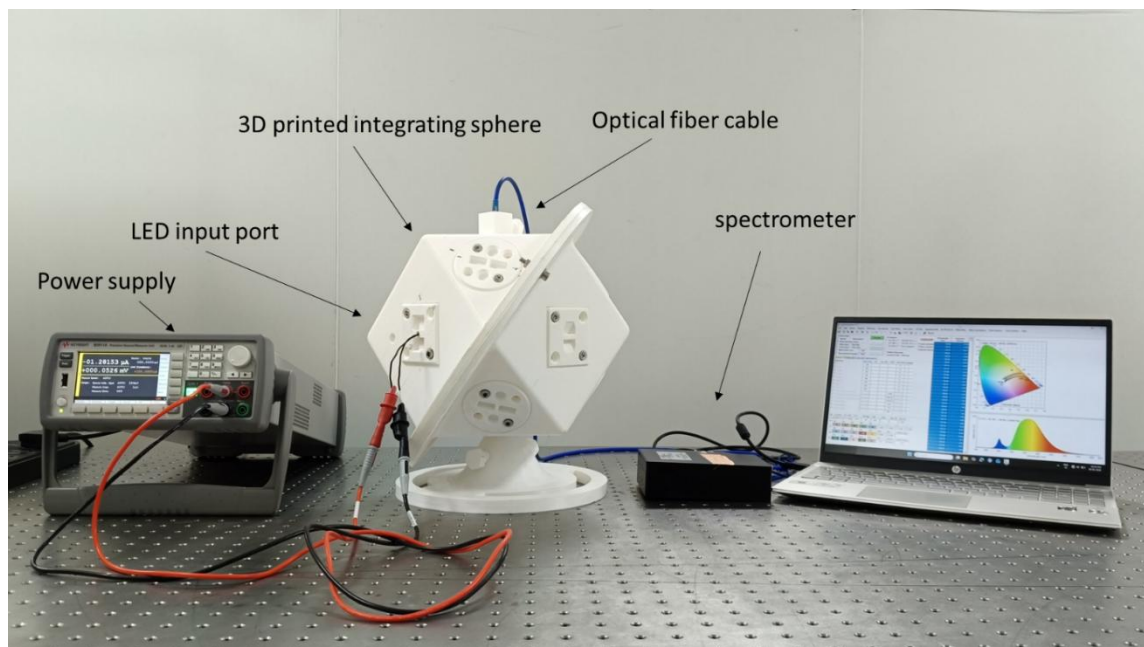
Expected Results:

- **Spectrum:** A distinct emission peak (e.g., ~450 nm for blue InGaN LEDs) with a measurable Full Width at Half Maximum (FWHM) and broad yellow emission together giving white light spectrum

Precautions:

- ✓ Never exceed the absolute maximum forward current rating of the LED, as self-heating will rapidly degrade the quantum efficiency and shift the emission wavelength (red-shift).
- ✓ Ensure the integrating sphere is completely closed and latched before taking baseline dark measurements or active sweeps to prevent external light contamination.

- ✓ Do not touch the interior reflective coating of the integrating sphere, as any degradation or smudging dramatically alters the sphere's multiplier and invalidates the calibration.



Experiment 3: FREE SPACE OPTICAL COMMUNICATION

Objective:

To establish a free space optical communication (FSOC) link through a fog chamber, characterize fog-induced attenuation, compare forward error correction (FEC) schemes, and evaluate bit error ratio (BER), encoded data size and transfer time under clear, light, mild and severe fog conditions.

Theory:

The transmitter converts file data into a coded binary sequence that drives a laser. The beam traverses the fog chamber and returns through a mirror assembly to the photodetector. Fog reduces received power through scattering and absorption. The receiver recovers and decodes the signal; XOR comparisons at corresponding processing stages quantify errors before and after decoding. FEC adds redundancy to correct some errors at the cost of increased data size and processing time.

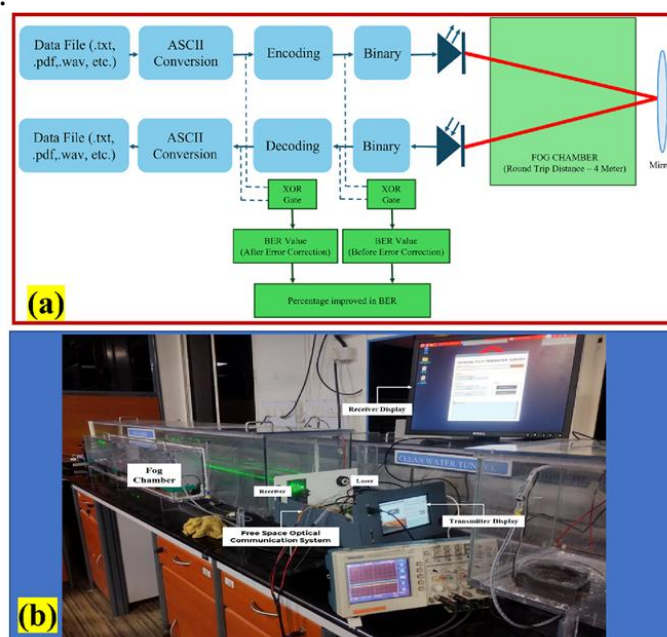


FIGURE 1. (A) FSOC SIGNAL FLOW AND 4 M OPTICAL PATH; (B) LABELLED EXPERIMENTAL SETUP [1].

Experimental Apparatus:

Apparatus	Configuration or purpose
FSOC transmitter and receiver	Two Raspberry Pi units with displays and file-transfer software; laser drive electronics; photodetector receiver and oscilloscope.
Optical source and channel	Green laser: 532 ± 10 nm, 100 mW reported; 2 m fog chamber with secured mirror assembly giving a 4 m folded path.
Fog generation and calibration	Fog generator with fan/heater controls; approved fog medium; optical power meter. Paper calibration used a nominal 5 mW red laser at 632 ± 10 nm over 2 m.

Supporting equipment	Regulated supplies, optical mounts, reference-file storage, beam barriers/stops and laser eyewear suitable for the installed sources.
----------------------	---

Equipment note: Confirm installed ratings, detector logic levels and FEC parameters before use. The green data laser and red calibration laser serve different purposes; their attenuation measurements are not interchangeable.

Procedure:

- 1. Establish the clear link:** With fog generation off and the chamber clear, inspect wiring and secure all optics. Align the outgoing and return paths at the lowest permitted power. Verify a stable receiver waveform and compatible controller input levels.
- 2. Verify file transfer:** Open the FSOC software on both Raspberry Pi units. Preserve a known reference file, select no FEC and matching settings, press Receive first, then Transfer. Save the received file, waveform, byte count and transfer time.
- 3. Calibrate fog:** Use a power sensor rated for the incident power and set for the calibration wavelength. Record the clear reference power over a fixed path, generate fog, switch the generator off and retain fan circulation. Record received power every 30 s and identify a stable measurement interval.
- 4. Set fog conditions:** The paper used generator ON for 1 min with fan level 3 for very light fog; 1 min with level 5 for mild fog; and 5 min with level 5 for severe fog. These are testbed-specific settings: confirm each condition from measured power and stability before transfer.
- 5. Compare FEC codes:** At each fog condition, test no FEC, repetition 3, repetition 5, Hamming (7,4), BCH and Reed–Solomon. The paper also discusses Turbo coding; include it only if implemented and verified. Record code parameters, encoded size, received data and elapsed time.
- 6. Repeat and reset:** Use the same payload, link rate, optical power and alignment; repeat each setting at least three times. Monitor fog drift and retain failed transfers. Disable emission before clearing the chamber through approved ventilation; restore the clear baseline between condition groups.



FIGURE 2. RECEIVED-BEAM APPEARANCE IN (A) MILD AND (B) SEVERE FOG [1].

Measurements and Calculations:

$BER = \text{Number of differing bits} / \text{Total compared bits}$, for equal-length, aligned files. State whether BER is measured before or after FEC decoding. Report missing or extra bytes separately; a shifted file comparison is not a reliable channel BER measurement. Record zero-error trials as “0 errors in N bits”.



Expected Results:

The paper reports repetition coding as useful in mild fog and Reed–Solomon/BCH as more effective under stronger fog. Verify these trends against measured overhead and errors. GUI modulation choices were future features in this study, not demonstrated modulation experiments. Exact BCH, RS and Turbo settings require verification in the installed software.

Precautions:

- ✓ Use trained laser supervision, appropriate eyewear, enclosed beam paths and beam stops; avoid direct and reflected exposure. The reported 100 mW source requires particular care. Keep hands and reflective objects out of both optical paths.
- ✓ Keep electronics dry; isolate power before rewiring. Use only approved fog fluid and ventilation, avoid inhalation and hot generator surfaces, and check sensor power limits. Stop for spills, overheating or uncontrolled beams; disable lasers before shutdown and chamber cleaning.

Experiment 4: Non-Invasive Haemoglobin Estimation Using Photoplethysmography and Machine Learning

Objective:

To acquire red and infrared photoplethysmography (PPG) signals using the MAX30102 interfaced with an Arduino Uno, record subject-wise data, validate the PPG waveform, process the signals using a Python-based digital signal processing pipeline, extract features, and demonstrate prototype-level haemoglobin estimation using XGBoost regression.

Theory:

Photoplethysmography measures pulsatile blood-volume changes in peripheral microvascular tissue. The PPG waveform contains a large quasi-static DC component and a smaller AC component that oscillates at the cardiac frequency. The MAX30102 provides red and infrared optical measurements. The optical interaction is influenced by wavelength-dependent absorption and scattering in tissue; the Modified Beer-Lambert Law accounts for the effect of tissue scattering.

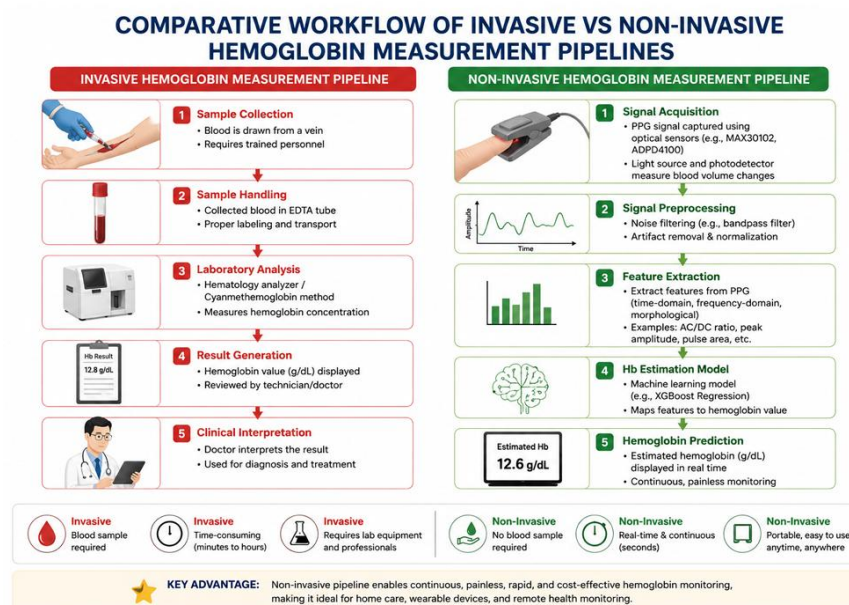


Figure 1. Comparative workflow of invasive and non-invasive haemoglobin measurement pipelines.

Experimental Apparatus:



Figure 2. Experimental hardware assembly used for optical data acquisition.

Component	Function / configuration described in thesis
MAX30102	Dual-wavelength optical sensing using red (660 nm) and IR (880 nm); integrated photodetector, analog conditioning, ADC and I2C interface.
Arduino Uno	ATmega328P embedded bridge; FIFO reads at 100 samples/s and UART streaming at 115200 baud.
I2C	MAX30102 via A4 (SDA) and A5 (SCL); address 0x57; 4.7 k Ω pull-ups to 3.3 V.
Python	Serial acquisition, metadata logging, CSV generation, waveform visualization, signal processing, feature extraction and model execution.
ADPD4100	Advanced optical AFE studied for future configurable multi-wavelength expansion; SPI and register-level configuration.

Procedure:

- Subject registration: Enter name, age, gender and session number in the Python host script.
- Sensor preparation: Clean the finger clip with an isopropyl alcohol wipe and ensure the subject has washed their hands.
- Finger placement: Place the index finger of the non-dominant hand into the sensor clip. Rest the finger comfortably without pressing excessively and maintain a constant position.
- Signal validation: Monitor the real-time serial output. Confirm that the IR value exceeds the finger-detection threshold and that a visible pulsatile waveform is present.
- Data recording: Start a 30-second recording. The subject remains still and breathes normally during acquisition.
- File saving: Save the CSV data file and waveform PNG generated by the acquisition/processing pipeline.

Signal-Quality and Fail-Safe Checks

The firmware and Python processing pipeline reject invalid recordings before they reach the machine-learning stage.

Condition	Detection criterion	System response
No finger	IR ADC < 50,000 counts	Prompt to place finger on sensor.

Non-living object	IR threshold met; AC amplitude < 500 counts	Reject as invalid non-physiological signal.
Flat signal	σ IR below minimum threshold after filtering	Signal quality failed; repeat measurement.
Insufficient data	Fewer than 200 valid rows in CSV	Discard recording and retry.
NaN / infinite features	Any feature = NaN or ∞	Reject invalid recording.

PPG Validation

- Waveform periodicity should be consistent with a resting heart rate of 50–90 BPM for seated subjects.
- At least 10 peaks should be detected in a 30-second recording.
- Heart-rate plausibility is checked within 40–180 BPM.
- Recordings failing any criterion are flagged for visual inspection and, if confirmed invalid, discarded.

Data Logging

- Each recording is saved as a structured CSV file containing a metadata header block with subject name, age, gender, recording timestamp and sensor configuration parameters. The column header is sample_index, red, ir, followed by one row per acquired sample containing the sample index and raw red/IR ADC values. The Python pipeline also generates a waveform PNG and an XGBoost-ready feature CSV.

Machine-Learning Procedure

- Collect valid subject CSV files and run the feature-extraction pipeline.
- Compile the labelled feature dataset; the thesis states that clinical training requires corresponding CBC Hb labels.
- Clean the feature data: NaN values are imputed using the column median and infinite values are clipped to the 1st/99th percentile range.
- Use an 80/20 stratified train-test split by Hb range bracket.
- Train the XGBoost regressor using the prototype configuration: n_estimators = 100, max_depth = 4, learning_rate = 0.1, subsample = 0.8, colsample_bytree = 0.8 and reg_lambda = 1.0.
- Evaluate using MAE, RMSE and R²; the trained model can be saved as a serialized .pkl file for deployment inference.

• Prototype Classification Logic

- For the prototype system: Anemic if predicted Hb < 10.0 g/dL; Normal if 10.0 ≤ predicted Hb ≤ 13.0 g/dL; High-Normal if predicted Hb > 13.0 g/dL. The thesis states that a clinically calibrated system would apply gender-specific and age-specific WHO Hb thresholds.

```

XGBoost Model Ready ✓

Raw Data Loaded:
  timestamp  ch660  ch730  ch850  ch940
0          1.75  2.727  3.75  4.765
1 21:29:27.432460 1.75  2.727  3.75  4.765
2 21:29:27.524475 1.75  2.727  3.75  4.765
3 21:29:27.605121 1.75  2.727  3.75  4.765
4 21:29:27.677802 1.75  2.727  3.75  4.765

Extracted Features:
  red_mean  ir_mean  red_std  ir_std  ratio_R
0  0.884809  3.721312  0.522185  0.332662  1.569478

===== FINAL RESULT USING XGBOOST =====
Estimated Hb: 14.63 g/dL
Condition: High Normal
  
```

• *Figure 3. Example prototype XGBoost Hb-estimation output.*

Precautions and Limitations

- Maintain repeatable finger placement and avoid excessive contact pressure; pressure changes can reduce fingertip perfusion and weaken the pulsatile AC component.
- Use the opaque, ambient-light-shielded finger-clip configuration described in the thesis and avoid direct sunlight; the reported shielding reduced baseline DC noise by approximately 30–40%.
- No-finger testing produced IR values below 10,000 raw counts, while a rubber eraser could exceed the IR threshold but lacked physiological AC variation; therefore, both IR intensity and pulse activity are required for validity.
- The present work is a research prototype. Clinical validity requires CBC-labelled calibration data; the thesis identifies approximately 100–200 diverse subjects and independent-cohort validation as the clinical calibration requirement.
- The prototype dataset used synthetically assigned Hb values distributed across 8–16 g/dL for proof-of-concept evaluation because true CBC Hb labels had not yet been collected in the clinical setting.

Experiment 5: Pulsed Laser Deposition (PLD): Demonstration & Hands-On Orientation

1. Objective of the Demonstration

The purpose of this demonstration is to introduce participants to the operation and working principle of a Pulsed Laser Deposition (PLD) system and to demonstrate how thin films can be deposited from a solid target onto a substrate.

At the end of the demonstration, participants should be able to:

- Explain the basic principle of PLD.
- Identify the major components of a PLD system.
- Understand the sequence of a typical PLD deposition.
- Understand the roles of laser energy, target, substrate, pressure, atmosphere and substrate temperature.
- Observe the initial stages of target preparation, chamber preparation and system setup.
- Understand how PLD parameters influence thin-film growth.
- Recognize the major safety precautions associated with laser, vacuum and high-temperature operation.

Important: Laser firing and operation of high-voltage/vacuum components will be performed only by the authorized system operator. Participants will observe these stages and will not independently operate the laser.

2. What is Pulsed Laser Deposition?

PLD is a physical vapor deposition technique used to prepare thin films and coatings.

Pulsed laser → Target → Laser ablation → Plasma plume → Substrate → Thin film

A high-energy laser pulse is focused onto a solid target inside a controlled vacuum or gas environment. The laser rapidly heats and ablates a small region of the target, producing a plasma plume containing atoms, ions, molecules and clusters. The ejected species travel toward the substrate and condense to form a thin film.

Basic Process Sequence:

Laser pulse → Target ablation → Plasma formation → Material transport → Film nucleation → Film growth

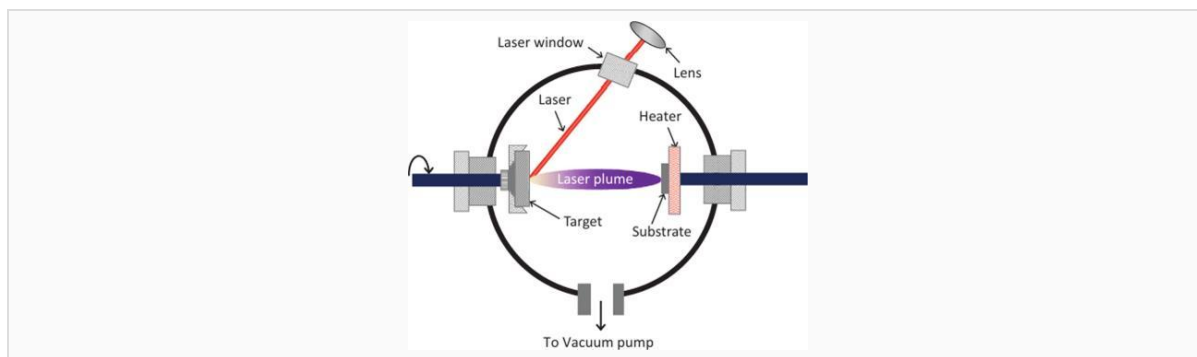
3. Major Components of a PLD System

Participants will be introduced to the following key system components:

Component	Function
Pulsed laser	Provides high-energy laser pulses for target ablation
Focusing optics	Focuses the laser beam onto the target
Vacuum chamber	Provides a controlled deposition environment
Target holder	Holds the material to be deposited

Substrate holder	Supports the substrate
Substrate heater	Controls substrate temperature
Vacuum pump system	Produces the required chamber pressure
Pressure gauge	Monitors chamber pressure
Gas inlet system	Introduces controlled gases such as O ₂ , Ar, etc.
Target rotation	Helps distribute laser irradiation over the target
Substrate rotation	Can improve film uniformity
Laser viewing/alignment	Helps monitor laser-target interaction
Control system	Controls and monitors deposition parameters

4. Typical PLD Chamber Configuration



Key Concept: The target is the source of material, while the substrate is the surface on which the thin film grows.

5. Demonstration Workflow

The complete PLD deposition sequence is executed in structured stages:

STEP 1 — Material & Target Preparation

The operator will demonstrate: Selection of deposition target • Checking target composition and condition • Mounting target securely • Checking target surface for contamination/damage • Setting target rotation.

STEP 2 — Substrate Preparation

Typical substrates: Silicon, Glass, Sapphire, Metal substrates, and other application-specific substrates.

Operator Sequence: Cleaning → Drying → Mounting → Loading

The substrate surface is extremely important because surface cleanliness, roughness and crystallinity strongly influence nucleation and film growth.

6. Loading the Chamber

The operator will demonstrate: Opening chamber according to system procedure → Installing target → Installing substrate → Checking target–substrate geometry → Closing chamber → Checking chamber sealing → Starting vacuum sequence.

Important Parameters: Key Geometrical Parameters: Target–substrate distance • Target orientation • Substrate orientation • Laser incidence angle • Plasma plume direction.
Transport Path: Target → Plasma plume → Substrate

(Note: The target–substrate distance directly affects transport of ablated species and influences film thickness and morphology.)

7. Vacuum & Gas Environment

After chamber closure, the chamber is evacuated using the vacuum system.

Sequence: Chamber closed → Pumping → Base pressure → Process gas introduction → Working pressure

Depending on the material, deposition may be performed under: High vacuum, Oxygen, Argon, Nitrogen, or other controlled atmospheres.

Why is pressure important? The background gas affects the movement and energy of ablated species. At different pressures: Plasma plume expansion changes • Particle scattering changes • Deposition rate can change • Film stoichiometry and microstructure can change.

8. Laser Alignment — Demonstration Only

The operator will explain laser alignment with respect to the target: Laser beam path identification • Focusing optics • Target position • Laser spot position • Target rotation • Optical alignment principles.

Participant Safety Rules:

Participants **MUST NOT**: Look into the laser beam • Look at target interaction region without approved viewing system • Adjust laser optics • Open laser interlocks • Operate laser control system • Enter restricted laser areas during operation.
Laser firing will be demonstrated by the trained operator **ONLY**.

9. Laser–Target Interaction & Observation Points

During deposition pulse: Laser pulse → Rapid energy absorption → Target heating/ablation → Plasma generation → Expansion of plasma plume → Transport toward substrate → Condensation → Thin-film growth.

Participants should observe plasma plume characteristics through the approved chamber viewing system.

Observation Checklist During Demonstration:

☐ Target position	☐ Laser spot position	☐ Plasma plume direction	☐ Plasma plume appearance
☐ Target rotation	☐ Substrate position	☐ Chamber pressure	☐ Substrate temperature

10. Important PLD Parameters

Parameter	What It Influences
Laser fluence	Ablation rate and plume characteristics
Pulse energy	Energy delivered to target
Pulse repetition rate	Deposition rate and film growth
Wavelength	Laser–material interaction
Target–substrate distance	Particle transport and thickness
Substrate temperature	Surface diffusion, crystallinity and morphology



Background pressure	Plume expansion and particle scattering
Gas composition	Film chemistry/stoichiometry
Deposition time / pulses	Film thickness
Target / Substrate rotation	Target utilization & film uniformity

11. What Participants Will Observe & Film Evaluation

Stages of Observation: Stage A (System ID) • Stage B (Pre-Deposition Setup) • Stage C (Operator Laser Firing & Plume) • Stage D (Film Characterization).

Post-Deposition Characterization Techniques: XRD (Crystal structure) • Raman spectroscopy (Vibrational properties) • SEM (Surface morphology) • AFM (Surface roughness) • UV–Vis (Optical properties) • Profilometry/Ellipsometry (Thickness).

14. Safety Rules & Completion Checklist

SAFETY RULES (MUST FOLLOW):

LASER: Do not look directly into laser beam, bypass interlocks, or enter restricted areas.

VACUUM: Do not open chamber under vacuum; follow approved venting.

HIGH TEMP: Do not touch heated components; allow cooling time.

ELECTRICAL: Do not open high-voltage panels.

Workshop Completion Activity (Participant: _____)	Status
Identified PLD system components	☐ Completed
Understood PLD principle & parameters	☐ Completed
Observed target & substrate loading	☐ Completed
Understood vacuum generation & pressure control	☐ Completed
Observed laser alignment & ablation / plasma plume	☐ Completed
Understood safety procedures	☐ Completed

Instructor/Operator Signature: _____ Date: _____

Remember: PLD is a controlled interaction between laser energy, target material, plasma dynamics, surface processes and thin-film growth.

Experiment 6: Laser Modulation Assisted QEPAS for Sensitive Gas Sensing Applications

1. Aim / Objective

To study Quartz-Enhanced Photoacoustic Spectroscopy (QEPAS) for trace gas detection and to acquire the photoacoustic signal generated by H_2S and NH_3 gases using an in-house designed Transimpedance Amplifier (TIA), Precision Rectifier based signal conditioning circuit and a single tunable laser source.

2. Theory

Photoacoustic spectroscopy is based on the generation of acoustic waves when modulated optical radiation is absorbed by gas molecules. The absorbed optical energy is converted into heat through vibrational-translational relaxation, producing periodic pressure variations in the surrounding gas.

In QEPAS, a quartz tuning fork (QTF) operating near its resonance frequency of 32.768 kHz is used as an acoustic transducer. The pressure waves generated due to gas absorption excite the QTF, which produces a piezoelectric current proportional to the gas concentration.

The photo-acoustic signal amplitude S_{PA} is expressed as:

$$S_{PA} \propto P \cdot \alpha \cdot C \cdot Q$$

where P is the optical power, α is the absorption coefficient per unit concentration, C is the gas concentration, and Q is the quality factor of the acoustic resonator.

The photoacoustic signal is proportional to laser power, absorption coefficient, gas concentration and Q-factor of the QTF.

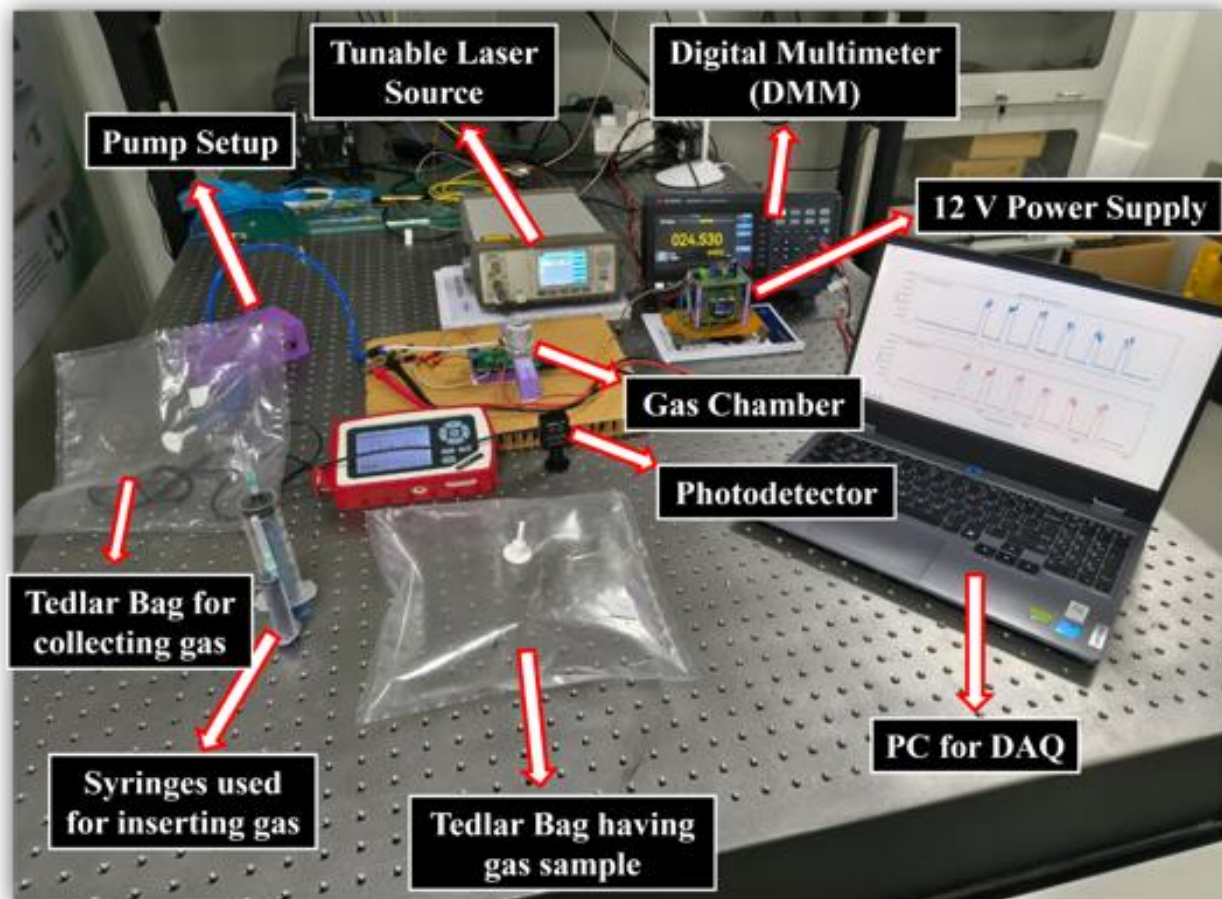
Since the QTF output current is extremely small, an in-house Transimpedance Amplifier (TIA) is used to convert the current signal into voltage. The amplified AC signal is then processed by a Precision Rectifier circuit which converts the low-level AC signal into a DC voltage proportional to the signal amplitude. The conditioned signal is subsequently digitized and displayed for analysis.

3. Experimental Procedure

- 1) Power ON the laser source, signal conditioning circuit and data acquisition system.
- 2) Align the optical beam through the acoustic detection region containing the quartz tuning fork.
- 3) Set the modulation frequency equal to the resonance frequency of the QTF.
- 4) Purge the chamber using nitrogen to establish the baseline signal.
- 5) Introduce H_2S gas into the chamber and allow the signal to stabilize.
- 6) Observe the amplified signal at the output of the TIA.
- 7) Record the corresponding Precision Rectifier output voltage.
- 8) Repeat the measurement for different H_2S concentrations.
- 9) Tune the laser to the NH_3 absorption wavelength and repeat the experiment.

10) Record and compare the responses obtained for NH_3 and H_2S gases.

Experimental Setup



4. Observations

Table 1: H_2S Gas Response

Concentration (ppm)	TIA Output (mV)	Rectifier Output (mV)

Table 2: NH_3 Gas Response

Concentration (ppm)	TIA Output (mV)	Rectifier Output (mV)



5. Discussion

The QTF generated a measurable piezoelectric signal when exposed to modulated optical absorption. The TIA provided sufficient amplification of the weak current signal while the Precision Rectifier enabled accurate extraction of the signal envelope. The output voltage increased with increasing gas concentration.

6. Result / Conclusion

QEPAS-based detection of H_2S and NH_3 gases was successfully demonstrated. The in-house designed TIA and Precision Rectifier circuit effectively amplified and conditioned the QTF signal for data acquisition. The observed output voltage showed a direct relationship with gas concentration, validating the signal conditioning architecture.

7. Precautions

- ✓ Do not touch the QTF prongs.
- ✓ Ensure proper grounding of the TIA circuit.
- ✓ Avoid mechanical vibrations near the sensor.
- ✓ Maintain stable laser power.
- ✓ Verify correct power supply polarity.
- ✓ Follow gas handling safety procedures.

Experiment 7: Underwater Optical Communication

1. Objectives

To establish a laser-based underwater optical link; compare SISO, SIMO, MISO and MIMO configurations; explain equal-gain combining (EGC); evaluate five modulation formats and forward error correction (FEC); and measure the effect of nominal 0, 1 and 2 NTU turbidity on bit error ratio (BER).

2. Learning Outcomes

The learner should be able to establish the link, compare spatial configurations, explain EGC, evaluate modulation and coding schemes, and assess the influence of turbidity on BER and link performance.

3. Principle

The transmitting computer encodes and modulates a file. USB-to-TTL interfaces and laser drivers with bias tees apply the data waveform and DC bias to the lasers. After propagation through water, photodetector branch voltages are combined with approximately equal weights. Signal conditioning, demodulation and decoding recover the file. All active lasers transmit the same information; spatial diversity adds optical paths, while FEC adds redundancy to correct some errors.

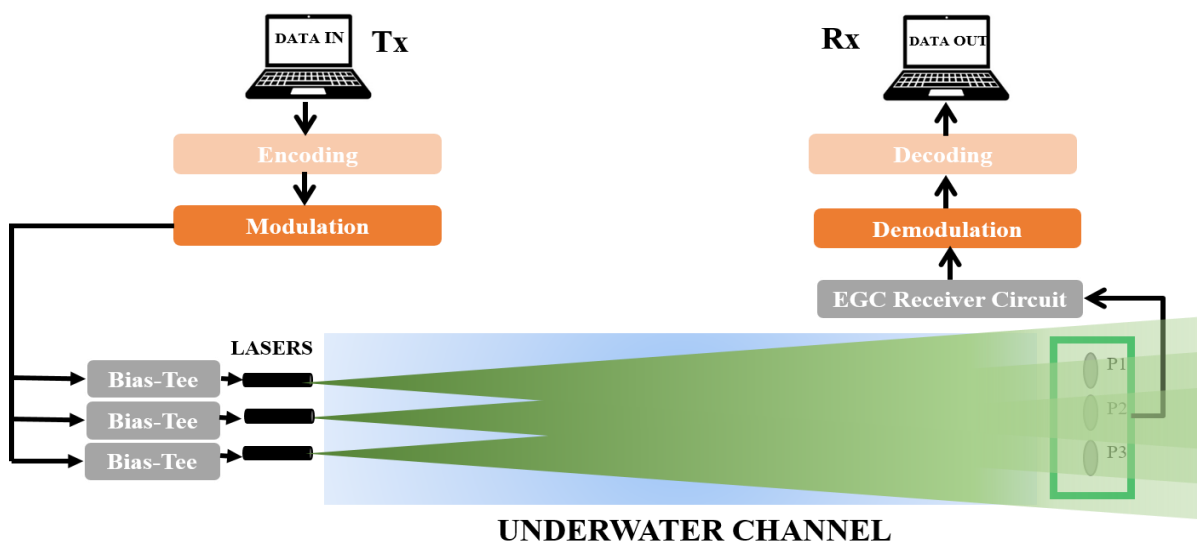


FIGURE 1. UWOC SYSTEM WITH SPATIAL DIVERSITY AND EGC RECEPTION.

4. Experimental Apparatus

Component	Function or configuration
Optical channel and branches	One-metre water tank; three laser transmitters with drivers and bias tees; three photodetector branches.
Receiver and computer interfaces	LM358 EGC receiver with verified logic conditioning; two USB-to-TTL adapters; two computers with the in-house GUI.
Measurement and safety equipment	Oscilloscope, probes, regulated supplies, multimeter, turbidity meter, approved turbidity medium, and suitable laser eyewear.

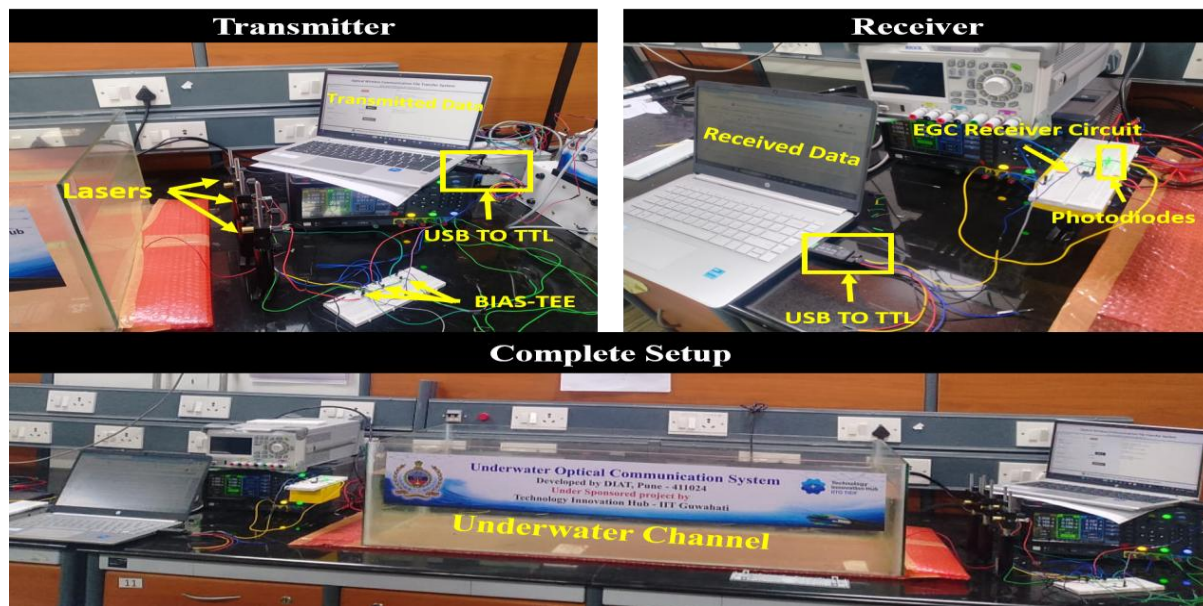


FIGURE 2. TRANSMITTER, RECEIVER AND COMPLETE UNDERWATER TESTBED.

5. Experimental Procedure

1. Preparation: Record laser ratings, driver limits, detector model, receiver supplies and adapter logic voltage. With power off, inspect wiring and ensure the tank and optical mounts are secure.
2. Receiver verification: Keep adapter RX disconnected. Check detector outputs, EGC output and conditioned signal using an oscilloscope. Correct clipping, polarity errors and out-of-range logic levels before connecting RX.
3. Baseline link: Measure water channel and actual baseline NTU. Select one laser and one detector (1×1). Align at the lowest permitted laser power and verify reception at the lowest supported rate using uncoded OOK-NRZ.
4. File transfer: Set matching COM-port, baud, framing, modulation and coding parameters at both computers. Start reception before transmission; send a known file and save the recovered file and waveforms. Repeat each setting at least three times.
5. Spatial diversity: With emission disabled during changes, test all nine configurations listed below. Keep the file, water condition, alignment and receiver gain controlled. Record whether per-laser power or total transmitted power is held fixed.
6. Modulation comparison: With FEC off, repeat for OOK-NRZ, OOK-RZ, PPM, DPIM and DHPIM at supported rate points. Match pulse mapping, symbol order, guard slots and header conventions. Record baud and measured optical timing separately.
7. Turbidity comparison: Use one fixed OOK format and no FEC. Prepare nominal 0, 1 and 2 NTU conditions with the approved medium, mix consistently and allow bubbles to clear. Measure actual NTU before and after trial groups; repeat the spatial configurations.
8. FEC comparison: At a fixed, recorded spatial configuration and water condition, compare no FEC with repetition 3, repetition 5, Hamming (7,4), BCH and Reed–Solomon for each modulation. Match code parameters and padding; record decoded-file errors, failures and transfer time.
9. Shutdown: Disable lasers, switch off drivers and supplies, disconnect USB, save all records and clean the tank only after electrical isolation.

6. Spatial Configurations

$M \times N$ denotes M active lasers and N active detectors. SISO: 1×1 ; SIMO: 1×2 and 1×3 ; MISO: 2×1 and 3×1 ; MIMO: 2×2 , 2×3 , 3×2 and 3×3 . Increasing transmitter count at



fixed per-laser power also increases total launch power; account for this when interpreting improvement.

7. Signal Validation and Data Logging

Check stable branch signals, and receiver output within the adapter's voltage limits, modulation, FEC Codes, actual NTU, baud, optical channel, optical-power limit, file lengths, errors, transfer time and waveform. Record waveforms, and incomplete transfers.

8. Calculations and Result

$BER = \text{Number of differing bits} / \text{Total compared bits}$, for equal-length, aligned files. State whether BER is measured before or after FEC decoding. Report missing or extra bytes separately; a shifted file comparison is not a reliable channel BER measurement. Record zero-error trials as "0 errors in N bits".

9. Safety and Precautions

- ✓ Use laser controls and eyewear selected for the installed wavelength and power. Terminate all beams, avoid direct and reflected exposure, remove reflective objects, and disable emission before alignment changes or branch switching.
- ✓ Keep electronics dry and away from spills. Switch off supplies and disconnect USB before rewiring; verify current limits, polarity and oscilloscope grounding. Never connect negative or excessive receiver voltage directly to adapter RX.
- ✓ Keep ambient light, water path and alignment consistent; use only the approved turbidity medium. Stop for spills, heating, abnormal current or uncontrolled beams. Follow laboratory emergency procedures for suspected exposure.

Experiment 8: Advanced Characterization of Ultrafast Fiber Laser Systems

1. Objectives

To familiarize the methods of characterization of an ultrafast fiber laser including autocorrelation, handling an optical spectrum analyzer, visualization of pulse traces, RF spectrum, etc.

2. Learning Outcomes

- Measure average output power and calculate pulse energy and peak-power estimates.
- Acquire and interpret optical spectra using an optical spectrum analyzer (OSA).
- Determine repetition rate and RF stability using a fast photodetector, oscilloscope and RF spectrum analyzer.
- Calculate time-bandwidth product (TBP) and assess whether pulses are transform-limited or chirped.
- Characterize wavelength drift, spectral stability and long-term power stability.
- Document measurements in a reproducible laboratory format and recognize common experimental artefacts

3. Experimental Procedure:

- Inspect the laser cavity, fiber connections, pump supply, cooling and enclosure.
- Verify that the output is routed through an appropriate isolator and/or attenuator before diagnostic instruments.
- Measure average power at low power first.
- Acquire the optical spectrum and record OSA settings.
- Detect the pulse train with a suitable fast photodetector and oscilloscope.
- Measure the repetition rate and RF spectrum.
- Measure pulse duration using the autocorrelator.
- Calculate pulse energy, approximate peak power and TBP.
- Repeat selected measurements at different pump powers and/or output coupling conditions.
- Perform a long-term stability measurement and document environmental conditions.

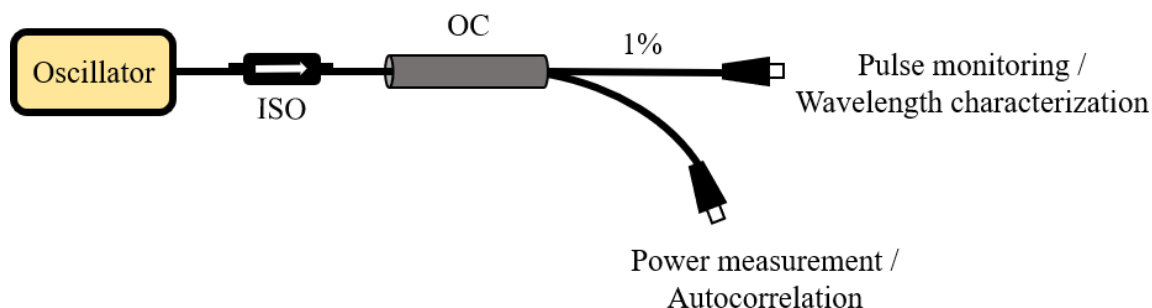


FIGURE 1. Apparatus for characterization of fiber laser oscillator

4. Safety and Precautions

- ✓ Wear wavelength-appropriate laser safety eyewear whenever required by the laboratory laser-safety assessment.



- ✓ Never look into a fiber connector, free-space beam, collimator, or microscope connected to an energized laser.
- ✓ Terminate unused fiber ports with appropriate power-rated terminations. Do not leave high-power outputs exposed.
- ✓ Use power-rated optical components. A component that survives the average power may still fail because of peak power, nonlinear effects or facet damage.
- ✓ Avoid unnecessary connector mating at high power. Inspect and clean fiber connectors before high-power operation.
- ✓ Use an optical isolator where back-reflection can destabilize the laser or damage the source.
- ✓ Reduce the laser to the minimum useful power before changing connections or measurement configurations.
- ✓ Do not exceed the damage threshold of photodiodes, autocorrelators, OSA inputs, or oscilloscope detectors.
- ✓ For 1 μm systems, check whether the chosen safety eyewear provides adequate protection at the actual wavelength. For 1.5 μm systems, remember that the beam may be invisible.
- ✓ Record emergency shutdown procedures and responsible laboratory personnel before beginning the experiment.

Experiment 9: Design and Characterization of Fiber Amplifiers

1. Objectives

Designing, constructing, testing and characterizing optical fiber amplifiers. It is intended for laboratory work on rare-earth-doped fiber amplifiers, especially Yb-doped amplifiers near 1 μm , Er-doped amplifiers near 1.5 μm , and Er/Yb co-doped amplifiers. The procedures are also applicable to amplifiers seeded by continuous-wave, quasi-CW, mode-locked, dissipative-soliton, dissipative-soliton-resonance and noise-like-pulse sources.

2. Learning Outcomes

- Understand the operating principle and architecture of a rare-earth-doped fiber amplifier.
- Select gain fiber, pump wavelength, pump direction, pump power, fiber length and passive components.
- Estimate small-signal gain, saturated gain, pump absorption and output power.
- Construct a safe fiber-amplifier test setup with appropriate isolation and attenuation.
- Measure gain, output power, optical efficiency, ASE and spectral evolution.
- Determine gain saturation and amplifier operating limits.
- Characterize pulse amplification without confusing gain with pulse distortion or nonlinear spectral broadening.
- Identify ASE, amplified spontaneous emission, self-phase modulation, stimulated Raman scattering and other limitations.
- Evaluate stability, thermal behavior and power scaling.
- Prepare a complete amplifier characterization report suitable for research publication.

3. Experimental Procedure

- Define the seed wavelength, seed power, pulse duration, repetition rate and spectral bandwidth.
- Define the required output power and desired gain.
- Select the gain fiber based on absorption, emission, mode-field diameter, numerical aperture and power-handling requirements.
- Select the pump wavelength and pump source power.
- Choose forward, backward or bidirectional pumping.
- Estimate the required doped-fiber length using pump absorption and gain requirements.
- Estimate passive loss and splice/connector loss.
- Check whether the intended peak power may cause self-phase modulation, SRS or other nonlinear effects.
- Select isolators, pump combiners, WDMs, mode-field adaptors and monitoring taps with adequate power ratings.
- Design the thermal-management arrangement.
- Construct and test first at low pump power.
- Increase pump power gradually while monitoring power, spectrum, ASE and temperature.

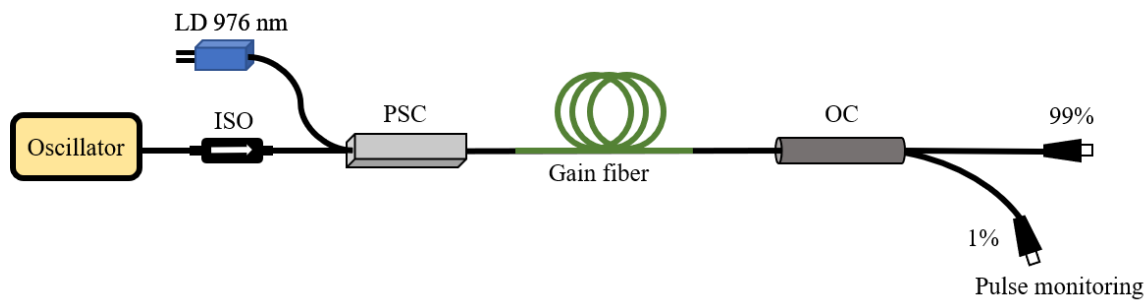


FIGURE 1. Schematics of fiber laser amplifier

4. Safety Requirements

- Fiber amplifiers can generate dangerous optical power, including invisible near-IR radiation and high peak powers. The amplifier must be treated as an energized laser system even when the seed power is low.
- Follow the laboratory laser-safety procedure and use wavelength-appropriate protective eyewear.
- Never inspect an energized fiber connector or output with the eye, even when no visible light is observed.
- Terminate unused outputs with components rated for the expected average and peak power.
- Use optical isolators where required to prevent reflections into the seed or gain stage.
- Do not exceed the maximum optical power of WDMs, combiners, isolators, photodiodes, OSA inputs or fiber connectors.
- Use calibrated attenuators before diagnostics whenever the expected power exceeds the instrument input limit.
- Allow high-power components and gain fibers to cool before handling them.
- For pulsed amplification, consider peak power as well as average power when assessing damage and nonlinear effects.
- Inspect and clean connectors before high-power operation.
- Use interlocks, beam enclosures and emergency shutdown procedures appropriate to the laboratory.

Experiment 10: Synthesis of $\text{Cs}_2\text{AgBiBr}_6$ Nanocrystals and Fabrication & Testing of a Paper-Based Photodetector.

A. Synthesis of $\text{Cs}_2\text{AgBiBr}_6$ (CABB) Nanocrystals

- **Aim:** To synthesize lead-free $\text{Cs}_2\text{AgBiBr}_6$ nanocrystals using a modified hot-injection method.
 - **Chemicals & apparatus:** $\text{Cs}(\text{OAc})$, $\text{Ag}(\text{OAc})$, $\text{Bi}(\text{OAc})$, oleylamine, oleic acid, octadecene, TMS-Br, n-hexane, centrifuge tubes, heating system, magnetic stirrer, nitrogen supply and centrifuge.
1. Dissolve the Cs, Ag and Bi precursors with oleylamine, oleic acid and octadecene.
 2. Heat the reaction mixture under vacuum to remove moisture and dissolved gases.
 3. Increase the temperature to the injection temperature under nitrogen atmosphere.
 4. Rapidly inject TMS-Br into the hot precursor solution.
 5. Allow the reaction to proceed for the specified reaction time.
 6. Immediately quench the reaction mixture in an ice-water bath.
 7. Centrifuge the product and collect the residue.
 8. Redisperse the residue in n-hexane and centrifuge again.
 9. Collect the supernatant containing the CABB nanocrystals

Parameter	Definition	Value
$\text{Cs}(\text{OAc})$	Cesium precursor amount	136 mg
$\text{Ag}(\text{OAc})$	Silver precursor amount	84 mg
$\text{Bi}(\text{OAc})$	Bismuth precursor amount	193 mg
Oleylamine	Surface-capping ligand	0.7 mL
Oleic acid	Surface-capping ligand	2.8 mL
Octadecene	Reaction solvent	10 mL
Vacuum treatment temperature	Temperature for precursor degassing	120 °C
Vacuum treatment time	Duration of degassing	30 min
Injection temperature	Temperature at which TMS-Br is injected	160 °C
TMS-Br	Bromide precursor	0.4 mL
Reaction time	Time after TMS-Br injection	10 s
Quenching	Rapid cooling medium	Ice-water bath
First centrifugation	Product separation condition	7000 rpm, 15 min
Redispersion solvent	Solvent used to redisperse residue	n-Hexane
Second centrifugation	Removal of unwanted precipitates	5000 rpm, 15 min

- **Observation:**

1. Colour of final dispersion: _____

2. Appearance/clarity: _____

3. Dispersion stability: _____

B. Fabrication and Testing of Paper-Based Photodetector

- **Aim:** To fabricate CABB and CABB/MoS₂ hybrid photodetectors on paper substrates and evaluate their photoresponse.
- **Materials:** CABB nanocrystal solution, MoS₂ nanoflowers, Whatman filter paper, silver electrodes and electrical measurement system.
- **Fabrication Procedure:**

1. Cut the Whatman filter paper into the required substrate size.
2. Immerse the paper substrate in the CABB nanocrystal solution.
3. Remove and dry the substrate; repeat the coating process to obtain a uniform film.
4. Anneal the coated substrate to improve film quality.
5. For hybrid devices, use CABB/MoS₂ nanoflower dispersion with the required MoS₂ composition.
6. Deposit silver electrodes on the coated paper.
7. Maintain the defined active area for electrical measurements.

Parameter	Definition	Value
Substrate	Paper used for device fabrication	Whatman filter paper
Substrate area	Geometrical area of paper used	0.5 × 0.5 cm ²
Coating cycles	Number of CABB deposition cycles	20
Annealing temperature	Temperature used to improve film quality	80 °C
Active area	Area used for photoresponse	0.5 × 0.5 cm ²
Electrode material	Conductive contact material	Silver (Ag)
Light source	Source used for illumination	Xe lamp
Illumination condition	Simulated solar illumination	43.9 mW/cm ²
Measurement instrument	Instrument for electrical measurement	Keithley 2420

• **Device Testing**

1. Connect the fabricated device to the source meter.
2. Record the dark I-V characteristics.
3. Illuminate the device using the calibrated light source.
4. Record the illuminated I-V characteristics.
5. Calculate the photocurrent from dark and illuminated currents.
6. Record photocurrent during repeated light ON/OFF cycles.
7. Determine responsivity, detectivity, response time and recovery time.

Parameter	Definition	Experimental value
Bias voltage	Applied voltage across the device	5 V
Dark current, I _(dark)	Current measured without illumination	

Light current, I_{light}	Current measured under illumination	
Photocurrent, I_{ph}	Difference between light and dark current	
Responsivity, R	Photocurrent generated per unit incident optical power	
Detectivity, D^*	Ability of detector to detect weak optical signals	
Rise time	Time required for photocurrent to reach illuminated level	
Recovery time	Time required for photocurrent to return toward dark level	

• **Observation Table:**

Device	I_{dark}	I_{light}	I_{ph}	R	D^*	Rise time	Recovery time
CABB							
CABB/MoS ₂ – Sample 1							
CABB/MoS ₂ – Sample 2							
CABB/MoS ₂ – Sample 3							

• **Calculations**

1. $I_{ph} = I_{light} - I_{dark}$

2. $R = \frac{I_{ph}}{P_{in}A}$

3. $D^* = \frac{R\sqrt{A}}{\sqrt{2eI_{dark}}}$

• **Result:**

- **Precautions:** Handle reactive precursors in a fume hood with appropriate PPE. Maintain the required atmosphere and temperature during synthesis. Avoid contamination of coated substrates and electrodes.



About the Organizers

	Defence Institute of Advanced Technology (DIAT) is the premier engineering training institute under the Department of Defence Research & Development, Ministry of Defence, Government of India. DIAT (DU) is specialized in the training of officers of Defence Research Organizations tri-services, IOFS (Indian Ordnance Factories), Defence PSUs (Like Hindustan Aeronautics Limited, Bharat Electronics, Bharat Dynamics Limited), ship building agencies like Garden Reach Shipbuilders & Engineers, Cochin and Goa Shipyards, Mazagon Dock Shipbuilders and armed forces of friendly countries (like Sri Lanka and Afghanistan,) and other central and state government agencies and civilians for their Masters and PhD. DIAT fosters collaboration, knowledge exchange, and a shared passion for photonics.
	OPTICA STUDENT CHAPTER: The DIAT Optica Student Chapter is an outreach initiative founded by the students of DIAT with the goal of connecting to the global community of optics and photonics researchers. The chapter aims to inspire and engage the broader public—especially students—by promoting awareness and understanding of optics. Serving as a bridge between DIAT and the wider community
	Sensors Research Society (SRS) Formed as a follow-up to the landmark conference on "Advances in Sensors for Aerospace Applications, Sensors 2007" held at Research Centre Imarat in December 2007. The society was established to synergise and accelerate the research and development of various sensors, driving the nation towards self-reliance in this critical technological domain.
	IITG-TIDF: Technology Innovation Hub (TIH), IIT Guwahati Technology Innovation and Development Foundation (TIDF) is an independent Section-8 company operating under the National Mission on Interdisciplinary Cyber Physical Systems. Operating under the Department of Science & Technology, Government of India, TIDF focuses on innovation and development.